

# ESTIMATION OF DETERMINISTIC PARAMETERS: LEAST SQUARES ESTIMATORS



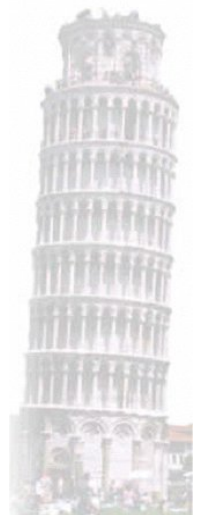
**Fulvio GINI**

Dipartimento di Ingegneria dell'Informazione

University of Pisa

Via G. Caruso 16, I-56122, Pisa, Italy

[fulvio.gini@unipi.it](mailto:fulvio.gini@unipi.it)



## Deterministic parameters:

probability of an event  $p = \Pr\{A\}$

moments of a distribution (mean value, variance, skewness, kurtosis, *etc.*)

parameters of a distribution

parameters of a signal (amplitude, phase, frequency, time of arrival, *etc.*)

moments of a random vector (mean vector, correlation matrix, *etc.*)

moments of a random signal (mean value function, ACF, PSD, *etc.*)

ACF=Autocorrelation Function, PSD=Power Spectral Density

## Deterministic parameters:

Sample estimates (application of the Law of large numbers)

Method of Moments (MM)

Maximum Likelihood (ML) method

Least Squares (LS): Linear LS or Non Linear LS (NLLS)



# The Least Squares (LS) Method

3

- The **Least Squares (LS)** method
- Math background: Solving linear equations
- **LS** solution for overdetermined linear equations: the LS estimator
- Math background:
  - The EigenValue Decomposition (EVD)
  - The Singular Value Decomposition (SVD)
- An example of application of **LS method**
- The **Deterministic Linear Model**: ML and LS estimators
- The **Best Linear Unbiased Estimator (BLUE)**
- **Sample estimates** of the **mean vector** and **covariance matrix** of a random vector

■ The **Least Squares (LS) estimator** of the deterministic vector  $\theta$  search for the value of  $\theta$  which minimizes the *distance* between the observed data vector and the signal of interest (SOI) vector model, i.e. the values that provide the best match between the observed data and the model:

$$\mathbf{x} = \mathbf{s}(\boldsymbol{\theta}) + \mathbf{e} \quad \Rightarrow \quad \hat{\boldsymbol{\theta}}_{LS} = \arg \min_{\boldsymbol{\theta}} \|\mathbf{e}\|^2 = \arg \min_{\boldsymbol{\theta}} \sum_{i=0}^{N-1} e_i^2$$
$$\hat{\boldsymbol{\theta}}_{LS} = \arg \min_{\boldsymbol{\theta}} \|\mathbf{x} - \mathbf{s}(\boldsymbol{\theta})\|^2 = \arg \min_{\boldsymbol{\theta}} \sum_{i=0}^{N-1} (x_i - s_i(\boldsymbol{\theta}))^2$$

■ **LS estimation** requires only knowledge of the signal model  $\mathbf{s}(\theta)$ .

■ If the signal vector depends linearly on the unknown parameters, then we call it **Linear Least Squares (LS)** estimator; if the dependency is not linear we call it **Non-Linear Least Squares (NLLS)** estimator.

- For example, in the case of a **sinusoid in noise**, the signal vector is a linear function only of the parameter  $A$ , but it is a nonlinear function of  $\theta$  and  $F_0$ :

$$\mathbf{s}(\boldsymbol{\theta}) \triangleq A \cdot \mathbf{p}(\theta, F_0) = \frac{A}{\sqrt{2B}} \begin{bmatrix} \cos(\theta) & \cos(2\pi F_0 + \theta) & \cdots & \cos(2\pi F_0 \cdot (N-1) + \theta) \end{bmatrix}^T$$

- As we will see in the following, in the **linear** case, the **LS estimator** is much easier to derive, it can always be expressed in closed form, and it is a linear function of the data.



# The Linear Least Squares (LS) Estimator

# Solving Linear Equations

7

- Assume we have to solve a system of  $N$  linear equations with  $L$  unknowns:

$$\sum_{i=1}^L a_{ki} x_i = b_k, \quad k = 1, 2, \dots, N$$


$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1L} \\ a_{21} & a_{22} & \cdots & a_{2L} \\ a_{31} & a_{32} & \cdots & a_{3L} \\ \vdots & \vdots & \ddots & \vdots \\ a_{N1} & a_{N2} & \cdots & a_{NL} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_L \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_N \end{bmatrix} \rightarrow \underset{N \times L}{\mathbf{A}} \underset{L \times 1}{\mathbf{x}} = \underset{N \times 1}{\mathbf{b}}$$

$$\underset{N \times L}{\mathbf{A}} \underset{L \times 1}{\mathbf{x}} = \underset{N \times 1}{\mathbf{b}}$$

**Problem:** Given  $\mathbf{A}$  and  $\mathbf{b}$ , determine  $\mathbf{x}$ .

$N$  number of equations,  $L$  number of unknowns

$N > L \rightarrow$  overdetermined system ( $\mathbf{A}$  is a "tall matrix")

$N < L \rightarrow$  underdetermined system ( $\mathbf{A}$  is a "fat matrix")

- If  $N < L$  there are more unknowns than equations, the linear system has  $\infty^{L-N}$  solutions.
- If  $N = L$  the number of equations is equal to the number of unknowns. In this case, if  $\mathbf{A}$  is full rank, i.e.  $\text{rank}(\mathbf{A}) = N$ , the system has only 1 solution:  $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$ .
- If  $N > L$  there are more equations than unknowns. In this case, if  $\mathbf{A}$  is full rank, i.e.  $\text{rank}(\mathbf{A}) = L$ , the system has a solution only if the vector  $\mathbf{b}$  belongs to the  $L$ -dim subspace generated by the  $L$  columns of  $\mathbf{A}$ :

$$\underset{N \times L}{\mathbf{A}} \underset{L \times 1}{\mathbf{x}} = \sum_{i=1}^L \underset{N \times 1}{\mathbf{a}_i} x_i = \underset{N \times 1}{\mathbf{b}}$$

$$\text{where } \mathbf{A} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_L]$$



■ *Example:*  $N=2, L=1, \mathbf{A}_{2 \times 1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{x}_{1 \times 1} = x$

If  $\mathbf{b}_{2 \times 1} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} \rightarrow \mathbf{b} \in R(\mathbf{A}) \rightarrow x = 3$

If  $\mathbf{b}_{2 \times 1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \rightarrow \mathbf{b} \notin R(\mathbf{A}) \rightarrow x = ?$  There is no solution!

$R(\mathbf{A})$  is the range of  $\mathbf{A}$ , i.e. the subspace spanned by the columns of  $\mathbf{A}$

## Least Squares (LS) solution for an overdetermined linear system of equations

■ Aim of the **Least Squares (LS)** solution:

Let us define:  $\underset{N \times 1}{\mathbf{e}} \triangleq \underset{N \times 1}{\mathbf{b}} - \underset{N \times L}{\mathbf{A}} \underset{L \times 1}{\mathbf{x}} \rightarrow \mathbf{b} = \mathbf{A}\mathbf{x} + \mathbf{e}, \text{ where } N > L$

$\mathbf{e} \equiv$  modeling error/measurement error

$$\mathbf{e} = [e_1 \quad e_2 \quad \cdots \quad e_N]^T \rightarrow \text{Error Energy: } E_e \triangleq \|\mathbf{e}\|^2 = \mathbf{e}^H \mathbf{e} = \sum_{i=1}^N |e_i|^2$$

**Problem:** find  $\mathbf{x}$  so that to minimize the energy of the error  $\|\mathbf{e}\|^2$

■ **Example:** let us consider an object that is moving of uniform rectilinear motion. Its position at time  $t$  is given by  $p(t)=\alpha+\beta t$ , with  $\alpha$  and  $\beta$  unknown parameters ( $\alpha$  is the initial position and  $\beta$  the speed).

■ To estimate the two unknown parameters, we can make two measurements of the object position at two different times. Obviously, the measurements will be affected by noise (e.g. internal noise of the measuring device):


$$b(t) = p(t) + e(t) = \alpha + \beta t + e(t)$$
$$\mathbf{b} = \begin{bmatrix} b(t_1) \\ b(t_2) \end{bmatrix} = \begin{bmatrix} \alpha + \beta t_1 + e(t_1) \\ \alpha + \beta t_2 + e(t_2) \end{bmatrix} = \begin{bmatrix} 1 & t_1 \\ 1 & t_2 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} + \begin{bmatrix} e(t_1) \\ e(t_2) \end{bmatrix} = \mathbf{A}\mathbf{x} + \mathbf{e}$$

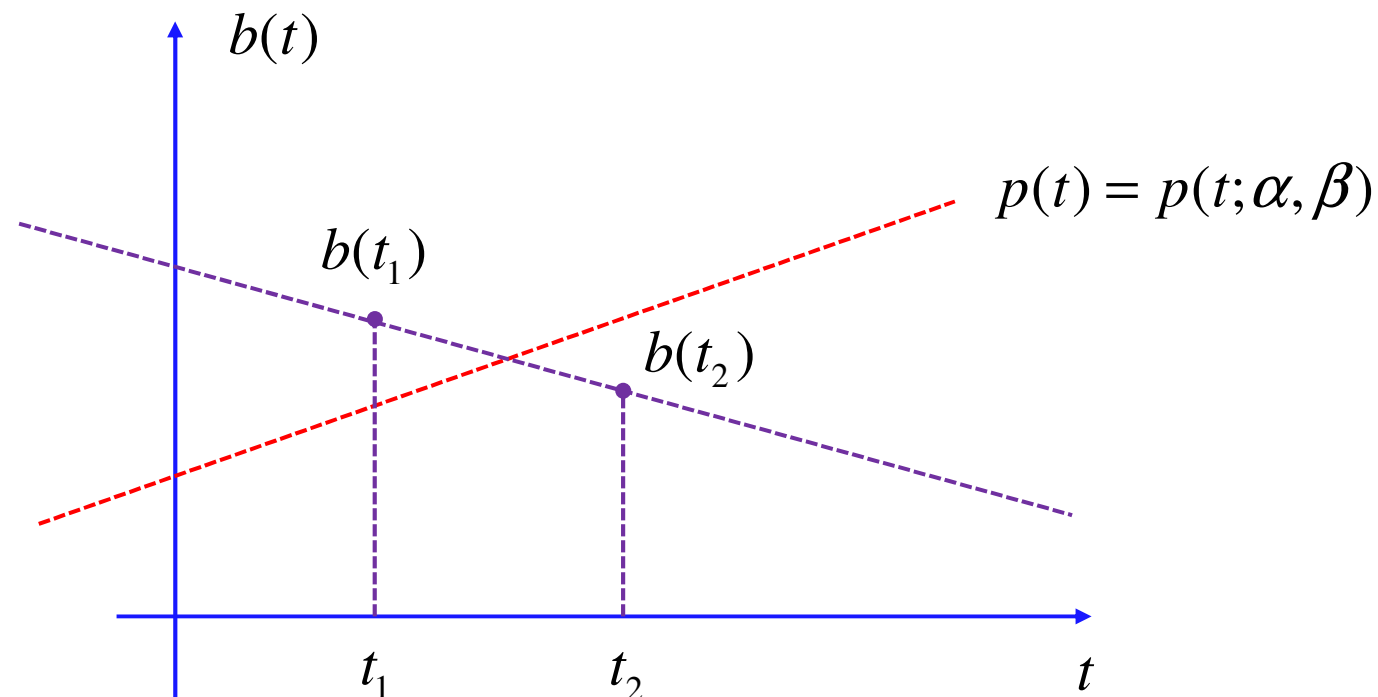
■ Matrix  $\mathbf{A}$  is known: it is determined by the kind of motion we hypothesize.



# Solving Linear Equations

12

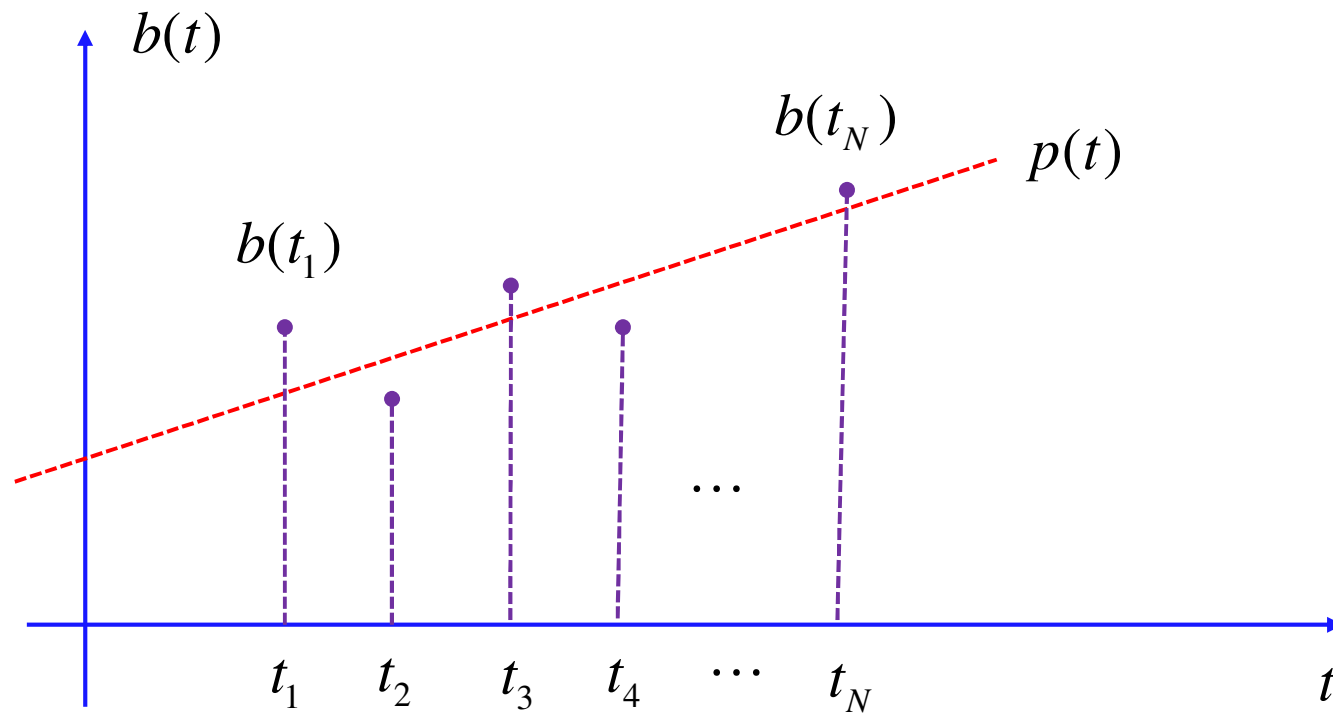
- If we solve the system of  $N=2$  linear equations in  $L=2$  unknowns, we obtain the parameters of the straight line that passes through the two positions we have measured:



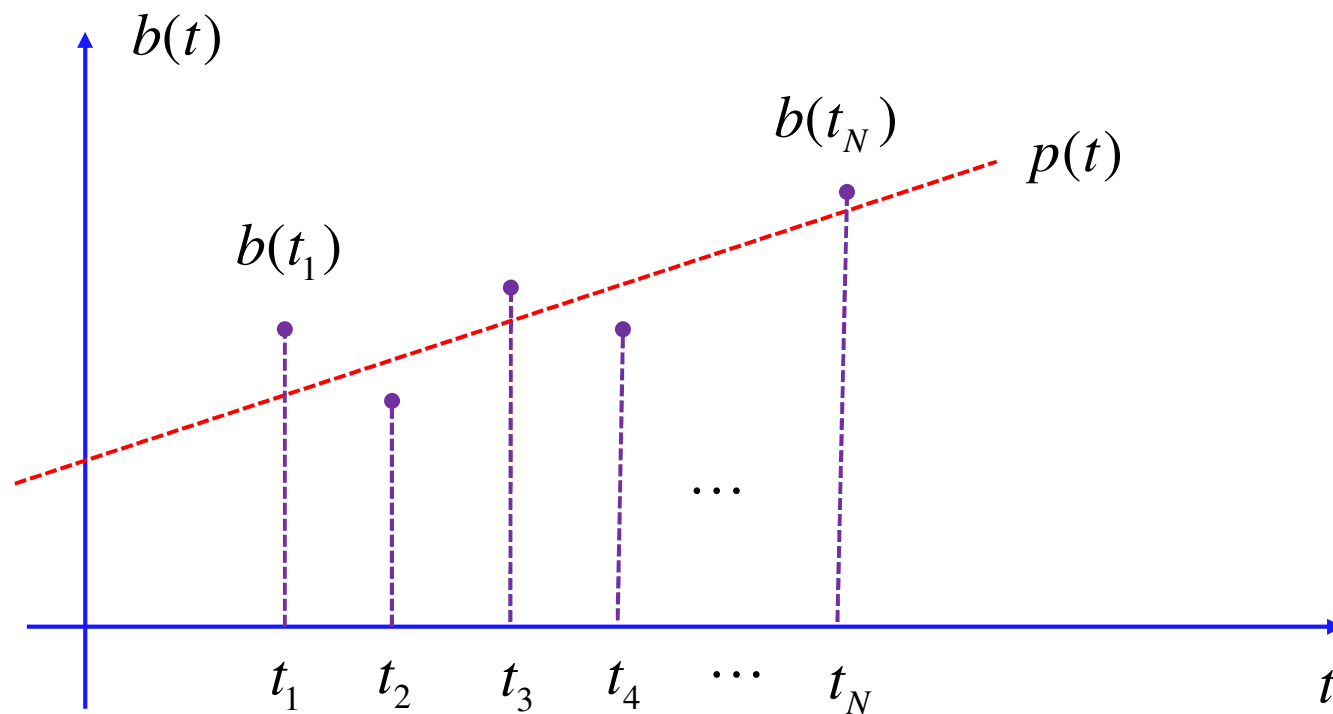
$$b(t_1) = p(t_1) + e(t_1), \quad b(t_2) = p(t_2) + e(t_2)$$

- To reduce (i.e. average out) the effect of the measurement noise, we should collect  $N > 2$  measurements:

$$b(t_i) = p(t_i) + e(t_i), \quad i = 1, 2, \dots, N$$

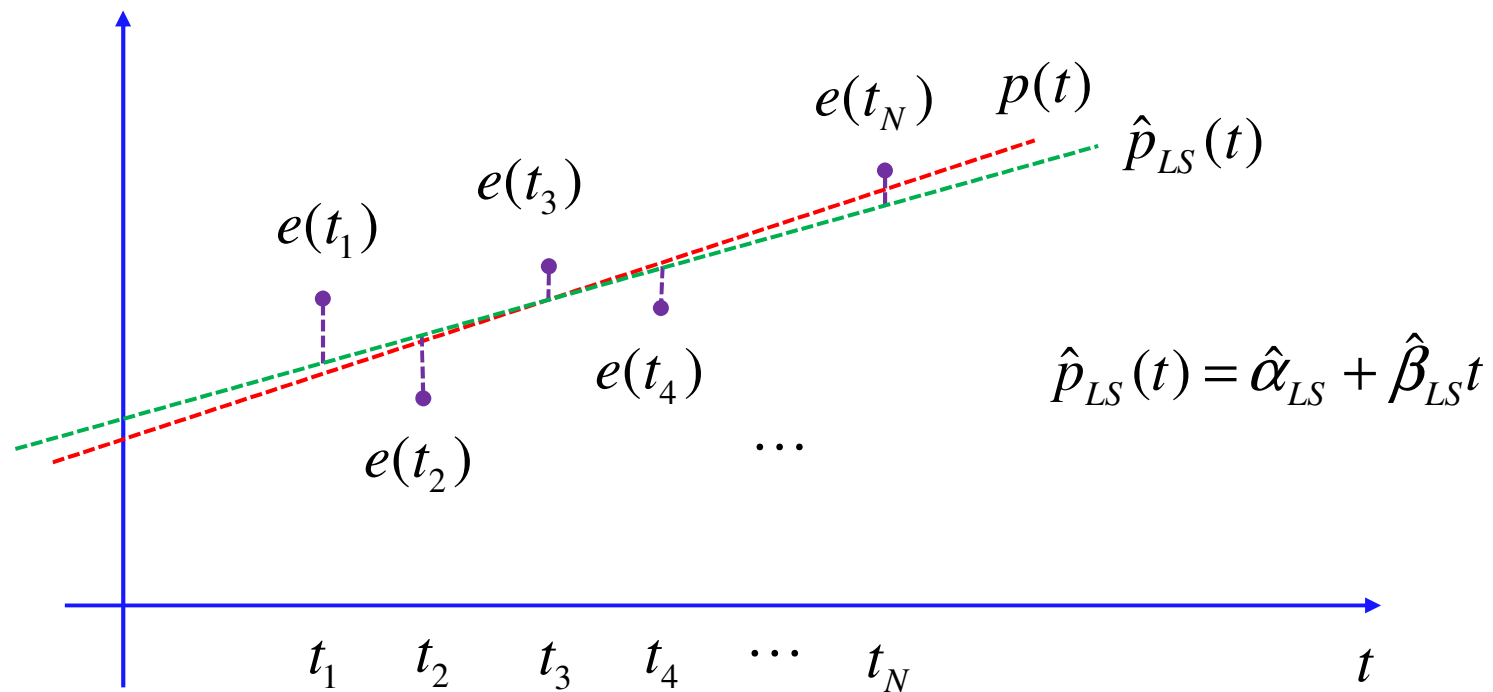


- $N$  measurements  $\rightarrow N$  equations and  $L=2$  unknowns.
- It does not exist a solution to this linear overdetermined system of equations, i.e. it does not exist a straight line that passes through all the  $N$  points:



- The idea of the **Least Squares (LS)** approach is to find the straight line that minimizes the energy of the errors between the measurements and the model (in this case: straight line motion). In other words, we look for the values of  $\alpha$  and  $\beta$  that minimizes the distance between model and measurements:

$$e(t_i) \triangleq e(t_i; \alpha, \beta) = b(t_i) - p(t_i; \alpha, \beta), \quad i = 1, 2, \dots, N$$



- The idea of the **Least Squares (LS)** approach is to find the straight line that minimizes the energy of the errors between the measurements and the model:


$$\mathbf{b} = \begin{bmatrix} b(t_1) \\ b(t_2) \\ \vdots \\ b(t_N) \end{bmatrix} = \begin{bmatrix} 1 & t_1 \\ 1 & t_2 \\ \vdots & \vdots \\ 1 & t_N \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} + \begin{bmatrix} e(t_1) \\ e(t_2) \\ \vdots \\ e(t_N) \end{bmatrix} = \mathbf{A}\mathbf{x} + \mathbf{e}$$

$$\hat{\mathbf{x}}_{LS} = \arg \min_{\mathbf{x}} \|\mathbf{e}\|^2 = \arg \min_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$$

$$\|\mathbf{e}\|^2 = \mathbf{e}^H \mathbf{e} = \sum_{i=1}^N |e_i|^2 = \sum_{i=1}^N |e(t_i)|^2$$





# Solving Linear Equations

17

$$\begin{aligned}\|\mathbf{e}\|^2 &= \mathbf{e}^H \mathbf{e} = (\mathbf{A}\mathbf{x} - \mathbf{b})^H (\mathbf{A}\mathbf{x} - \mathbf{b}) \\ &= \mathbf{x}^H \mathbf{A}^H \mathbf{A} \mathbf{x} - \mathbf{b}^H \mathbf{A} \mathbf{x} - \mathbf{x}^H \mathbf{A}^H \mathbf{b} + \mathbf{b}^H \mathbf{b} \\ &= \mathbf{x}^H (\mathbf{A}^H \mathbf{A}) \mathbf{x} - \mathbf{b}^H \mathbf{A} (\mathbf{A}^H \mathbf{A})^{-1} (\mathbf{A}^H \mathbf{A}) \mathbf{x} \\ &\quad - \mathbf{x}^H (\mathbf{A}^H \mathbf{A}) (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \mathbf{b} + \mathbf{b}^H \mathbf{b} \\ &= \left[ \mathbf{x} - (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \mathbf{b} \right]^H (\mathbf{A}^H \mathbf{A}) \left[ \mathbf{x} - (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \mathbf{b} \right] + \mathbf{b}^H \mathbf{b} \\ &\quad - \mathbf{b}^H \mathbf{A} (\mathbf{A}^H \mathbf{A})^{-1} (\mathbf{A}^H \mathbf{A}) (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \mathbf{b} \\ &= \left[ \mathbf{x} - (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \mathbf{b} \right]^H (\mathbf{A}^H \mathbf{A}) \left[ \mathbf{x} - (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \mathbf{b} \right] + \mathbf{b}^H \mathbf{b} \\ &\quad - \mathbf{b}^H \mathbf{A} (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \mathbf{b} \quad [\text{we assume here that } \mathbf{A} \text{ is full rank].}\end{aligned}$$

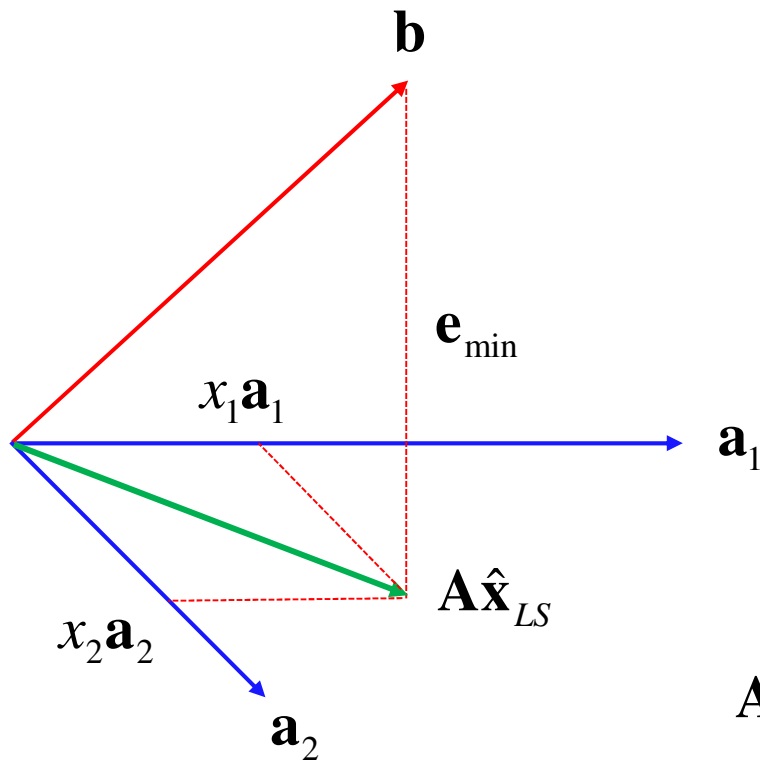
$$\begin{aligned}\|\mathbf{e}\|^2 &= \left[ \mathbf{x} - (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \mathbf{b} \right]^H (\mathbf{A}^H \mathbf{A}) \left[ \mathbf{x} - (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \mathbf{b} \right] \\ &\quad + \mathbf{b}^H \mathbf{b} - \mathbf{b}^H \mathbf{A} (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \mathbf{b} \\ &= \left[ \mathbf{x} - \mathbf{A}^\# \mathbf{b} \right]^H (\mathbf{A}^H \mathbf{A}) \left[ \mathbf{x} - \mathbf{A}^\# \mathbf{b} \right] + \mathbf{b}^H (\mathbf{I} - \mathbf{A} \mathbf{A}^\#) \mathbf{b}\end{aligned}$$

■ The 1<sup>st</sup> term is always  $\geq 0$ , whereas the 2<sup>nd</sup> term does not depend on  $\mathbf{x}$ . Hence, the energy of the error is minimized when the 1<sup>st</sup> term is 0:

$$\hat{\mathbf{x}}_{LS} = \mathbf{A}^\# \mathbf{b} = (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \mathbf{b} \quad \text{Least Squares (LS) solution}$$

$$\mathbf{A}^\# \triangleq (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \quad \text{Moore-Penrose pseudoinverse} \quad \rightarrow \quad \underset{L \times N}{\mathbf{A}^\#} \underset{N \times L}{\mathbf{A}} = \underset{L \times L}{\mathbf{I}}$$

- Geometric interpretation of the **linear Least Squares (LS)** solution:



Define:  $\mathbf{A}_{N \times 2} = [\mathbf{a}_1 \quad \mathbf{a}_2]$

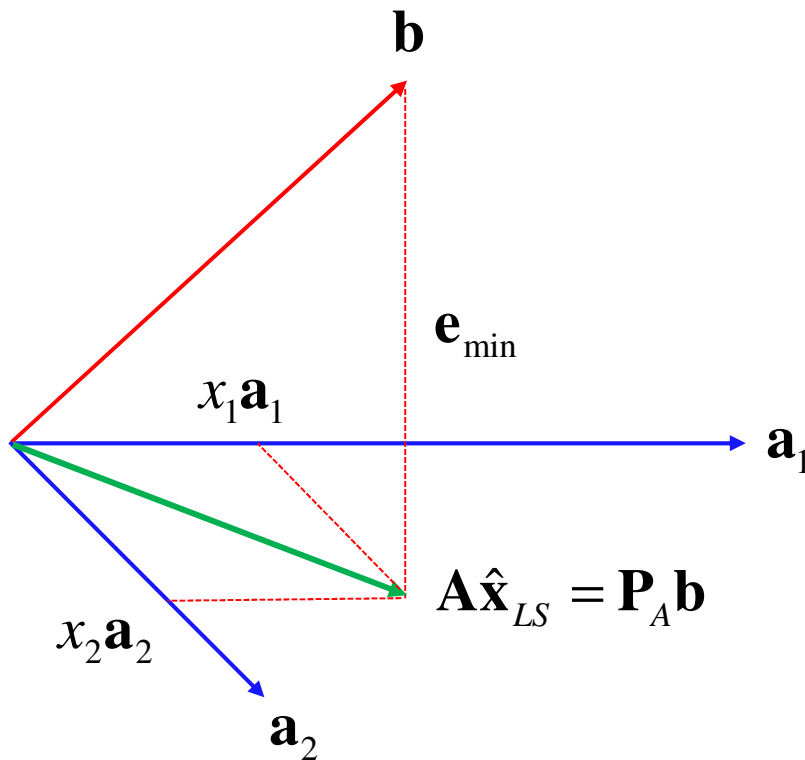
$$\hat{\mathbf{x}}_{LS} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{A}^\# \mathbf{b}$$

$$\mathbf{A}\hat{\mathbf{x}}_{LS} = \mathbf{A}(\mathbf{A}^\# \mathbf{b}) = \mathbf{A}(\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \mathbf{b} = \mathbf{P}_A \mathbf{b}$$

$$\mathbf{P}_A \triangleq \mathbf{A}\mathbf{A}^\# \text{ Projection matrix}$$

Project  $\mathbf{b}$  onto the subspace generated by the columns of  $\mathbf{A}$

- Geometric interpretation of the **linear Least Squares (LS)** solution:



$\mathbf{P}_A \triangleq \mathbf{A}\mathbf{A}^H$  Projection matrix

Property 1:  $\mathbf{P}_A = \mathbf{P}_A^2$  (Idempotent)

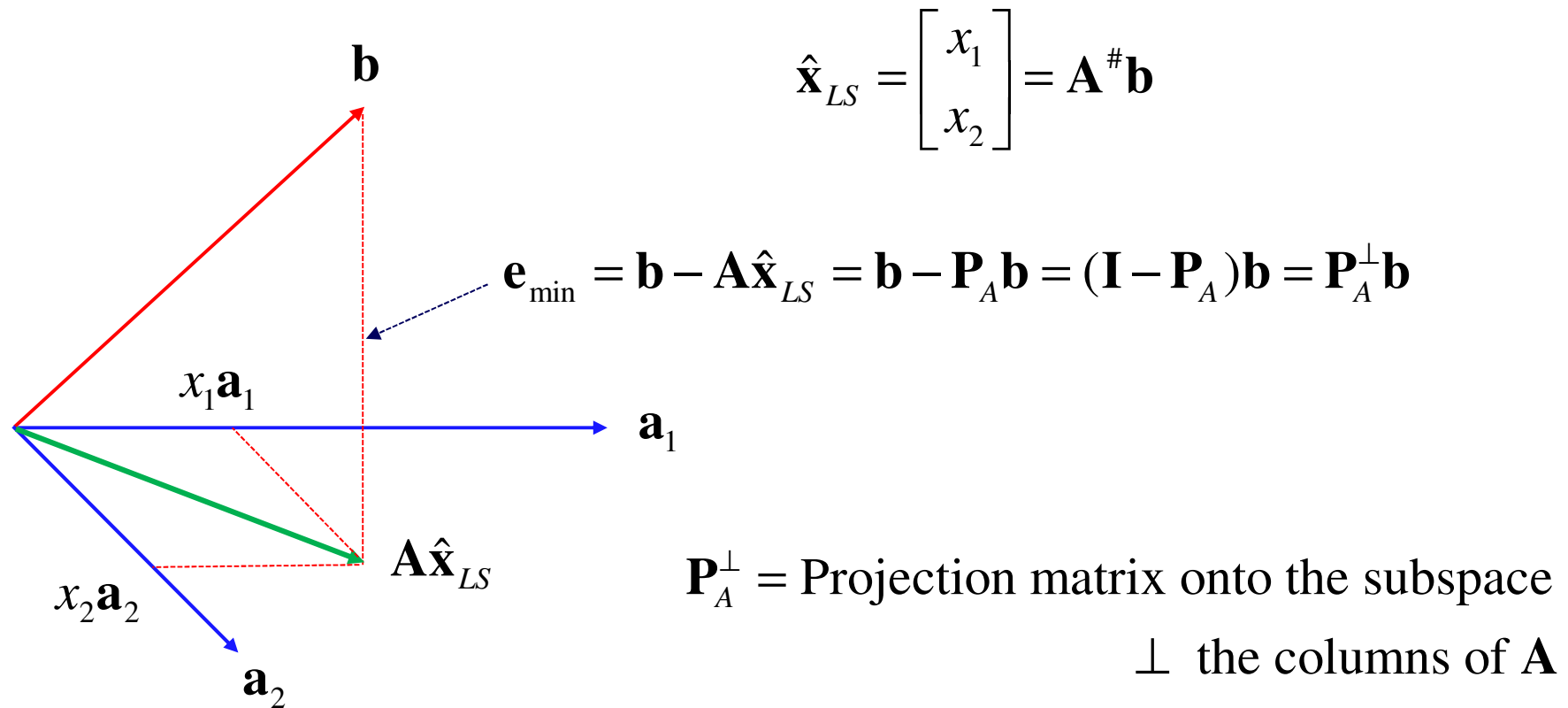
Property 2:  $\mathbf{P}_A = \mathbf{P}_A^H$  (Hermitian)

$$\text{P.1} \Rightarrow \mathbf{P}_A (\mathbf{P}_A \mathbf{b}) = \mathbf{P}_A^2 \mathbf{b} = \mathbf{P}_A \mathbf{b}$$

$$\text{P.1: } \mathbf{P}_A^2 = \mathbf{A}(\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \mathbf{A}(\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H = \mathbf{A}(\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H = \mathbf{P}_A$$

$$\text{P.2: } \mathbf{P}_A^H = \left( \mathbf{A}(\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \right)^H = \left( \mathbf{A}^H \right)^H \left( (\mathbf{A}^H \mathbf{A})^{-1} \right)^H \mathbf{A}^H = \mathbf{A}(\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H = \mathbf{P}_A$$

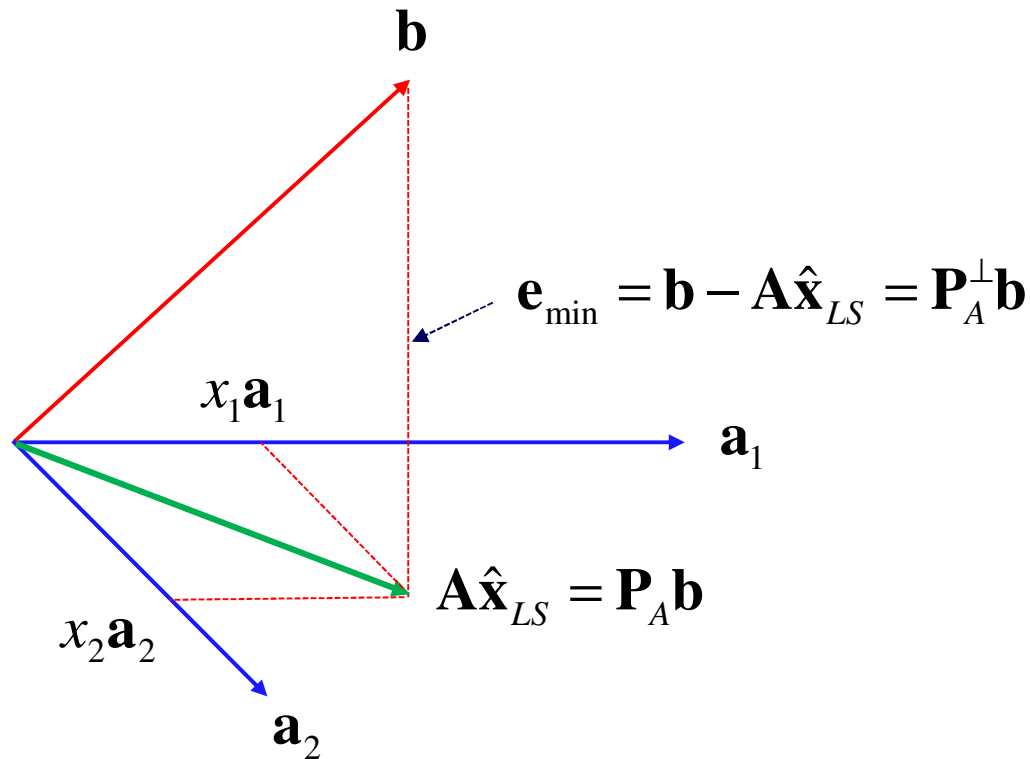
- Geometric interpretation of the **linear Least Squares (LS)** solution:



$$\mathbf{P}_A = \mathbf{A} \mathbf{A}^\# \neq \mathbf{A}^\# \mathbf{A} = \mathbf{I}$$

$\begin{matrix} N \times N & N \times L & L \times N & L \times N & N \times L & L \times L \end{matrix}$

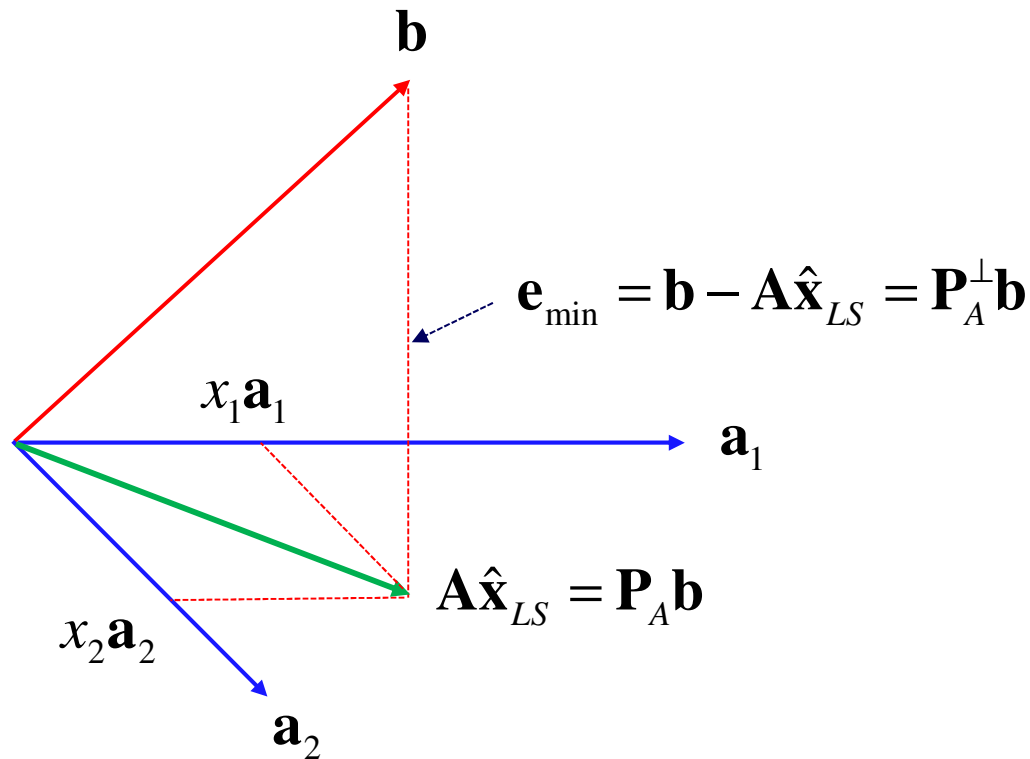
- Geometric interpretation of the **linear Least Squares (LS)** solution:



$$\hat{\mathbf{x}}_{LS} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{A}^\# \mathbf{b}$$

$$\begin{aligned} \|\mathbf{e}_{\min}\|^2 &= \mathbf{e}_{\min}^H \mathbf{e}_{\min} \\ &= (\mathbf{P}_A^\perp \mathbf{b})^H \mathbf{P}_A^\perp \mathbf{b} \\ &= \mathbf{b}^H (\mathbf{P}_A^\perp)^H \mathbf{P}_A^\perp \mathbf{b} \\ &= \mathbf{b}^H \mathbf{P}_A^\perp \mathbf{P}_A^\perp \mathbf{b} \\ &= \mathbf{b}^H \mathbf{P}_A^\perp \mathbf{b} \\ &= \mathbf{b}^H (\mathbf{I} - \mathbf{A} \mathbf{A}^\#) \mathbf{b} \end{aligned}$$

- Geometric interpretation of the **linear Least Squares (LS)** solution:



$$\hat{\mathbf{x}}_{LS} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{A}^\# \mathbf{b}$$

$$\mathbf{e}_{\min} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}}_{LS} \perp R(\mathbf{A})$$

$$\mathbf{e}_{\min} \perp \mathbf{a}_i \text{ for } i = 1, 2, \dots, L$$

- The linear LS solution is the vector  $\mathbf{x}$  that makes the error  $\mathbf{e}$  orthogonal to the range of  $\mathbf{A}$ , i.e. the subspace spanned by the columns of  $\mathbf{A}$ : **Orthogonality Principle** (a.k.a. **Projection Theorem**).

■ *Example:*  $N=2, L=1$ ,  $\mathbf{A}_{2 \times 1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ ,  $\mathbf{b}_{2 \times 1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ ,  $\hat{\mathbf{x}}_{LS} = ?$

$$\begin{aligned}\hat{\mathbf{x}}_{LS} &= \mathbf{A}^{\#} \mathbf{b} = \left( \mathbf{A}^H \mathbf{A} \right)^{-1} \mathbf{A}^H \mathbf{b} \\ &= \left( \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1\end{aligned}$$

$$\mathbf{A} \hat{\mathbf{x}}_{LS} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} (1) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\mathbf{e}_{LS} = \mathbf{b} - \mathbf{A} \hat{\mathbf{x}}_{LS} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \mathbf{e}_{LS} \perp \mathbf{A} \hat{\mathbf{x}}_{LS}$$



- **EigenValue Decomposition (EVD)** of an Hermitian square matrix:

$$\underset{N \times N}{\mathbf{A}} = \underset{N \times N}{\mathbf{U}} \underset{N \times N}{\mathbf{\Lambda}} \underset{N \times N}{\mathbf{U}^H} = \sum_{i=1}^N \lambda_i \mathbf{u}_i \mathbf{u}_i^H$$

$$\mathbf{U} = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \cdots \quad \mathbf{u}_N], \text{ unitary matrix of the eigenvectors} \quad \mathbf{U}^{-1} = \mathbf{U}^H$$

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_N \end{bmatrix}, \text{ diagonal matrix of the eigenvalues (all real)}$$

- If  $\mathbf{A}$  is full rank, i.e. all  $\lambda_i \neq 0$ , then we can calculate the inverse of  $\mathbf{A}$  as:

$$\mathbf{A}^{-1} = (\mathbf{U} \mathbf{\Lambda} \mathbf{U}^H)^{-1} = \mathbf{U} \mathbf{\Lambda}^{-1} \mathbf{U}^H$$

with Matlab:  
`eig.m`, `inv.m`

- The **Singular Value Decomposition (SVD)** is a generalization of the EVD for rectangular matrices:

$$\mathbf{A}_{N \times L} = \mathbf{U}_{N \times N} \mathbf{\Sigma}_{N \times L} \mathbf{V}_{L \times L}^H = \sum_{i=1}^L \sigma_i \mathbf{u}_i \mathbf{v}_i^H$$

$\mathbf{U} = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \cdots \quad \mathbf{u}_N]$ , unitary matrix of the left singular vectors

$\mathbf{V} = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_L]$ , unitary matrix of the right singular vectors

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_L \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}, \text{ matrix of the singular values (real, } \geq 0)$$

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_L \geq 0$$

- Hypothesis:  $N > L$  (tall matrix)

- If matrix  $\mathbf{A}$  is not full-rank, for example:  $\text{rank}(\mathbf{A}) = r < L$   
then, we have:


$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0 \quad \text{and} \quad \sigma_{r+1} = \sigma_{r+2} = \dots = \sigma_L = 0$$

$$\mathbf{U} = [\mathbf{U}_1 \quad \mathbf{U}_2], \quad \text{where} \quad \mathbf{U}_1 = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \dots \quad \mathbf{u}_r], \mathbf{U}_2 = [\mathbf{u}_{r+1} \quad \dots \quad \mathbf{u}_N],$$

$$\mathbf{V} = [\mathbf{V}_1 \quad \mathbf{V}_2], \quad \text{where} \quad \mathbf{V}_1 = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \dots \quad \mathbf{v}_r], \mathbf{V}_2 = [\mathbf{v}_{r+1} \quad \dots \quad \mathbf{v}_L],$$

$$\mathbf{\Sigma} = \begin{bmatrix} \mathbf{\Sigma}_1 & 0 \\ 0 & 0 \end{bmatrix}, \quad \text{where} \quad \mathbf{\Sigma}_1 = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_r \end{bmatrix}$$

- Matrix **A** not full-rank:  $\text{rank}(\mathbf{A}) = r < L$


$$\mathbf{U} = [\mathbf{U}_1 \quad \mathbf{U}_2], \quad \mathbf{V} = [\mathbf{V}_1 \quad \mathbf{V}_2], \quad \mathbf{\Sigma} = \begin{bmatrix} \mathbf{\Sigma}_1 & 0 \\ 0 & 0 \end{bmatrix},$$

$$\underset{N \times L}{\mathbf{A}} = \underset{N \times N}{\mathbf{U}} \underset{N \times L}{\mathbf{\Sigma}} \underset{L \times L}{\mathbf{V}^H}$$

$$= [\mathbf{U}_1 \quad \mathbf{U}_2] \begin{bmatrix} \mathbf{\Sigma}_1 & 0 \\ 0 & 0 \end{bmatrix} [\mathbf{V}_1 \quad \mathbf{V}_2]^H = [\mathbf{U}_1 \quad \mathbf{U}_2] \begin{bmatrix} \mathbf{\Sigma}_1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{V}_1^H \\ \mathbf{V}_2^H \end{bmatrix}$$

$$= [\mathbf{U}_1 \mathbf{\Sigma}_1 \quad 0] \begin{bmatrix} \mathbf{V}_1^H \\ \mathbf{V}_2^H \end{bmatrix} = \underset{N \times r}{\mathbf{U}_1} \underset{r \times r}{\mathbf{\Sigma}_1} \underset{r \times L}{\mathbf{V}_1^H}$$

$$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^H = \mathbf{U}_1 \mathbf{\Sigma}_1 \mathbf{V}_1^H$$

$\begin{matrix} N \times L & N \times N & N \times L & L \times L & N \times r & r \times r & r \times L \end{matrix}$

with Matlab:  
svd.m, pinv.m

- Efficient calculation of the pseudoinverse of  $\mathbf{A}$  is based on the SVD.
- Assume first that  $\mathbf{A}$  is full rank, i.e.  $r = \min(L, N) = L$ :

$$\mathbf{A} = \mathbf{U}_1 \mathbf{\Sigma}_1 \mathbf{V}_1^H, \quad \text{where } \mathbf{V}_1^{-1} = \mathbf{V}_1^H$$

$\begin{matrix} N \times L & N \times L & L \times L & L \times L \end{matrix}$

$$\mathbf{A}^\# = (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H = (\mathbf{V}_1 \mathbf{\Sigma}_1 \mathbf{U}_1^H \mathbf{U}_1 \mathbf{\Sigma}_1 \mathbf{V}_1^H)^{-1} \mathbf{V}_1 \mathbf{\Sigma}_1 \mathbf{U}_1^H$$

$L \times N$

$$= (\mathbf{V}_1 \mathbf{\Sigma}_1^2 \mathbf{V}_1^H)^{-1} \mathbf{V}_1 \mathbf{\Sigma}_1 \mathbf{U}_1^H = \mathbf{V}_1 \mathbf{\Sigma}_1^{-2} \mathbf{V}_1^H \mathbf{V}_1 \mathbf{\Sigma}_1 \mathbf{U}_1^H = \mathbf{V}_1 \mathbf{\Sigma}_1^{-1} \mathbf{U}_1^H$$

$$\mathbf{A}^\# \mathbf{A} = \mathbf{V}_1 \mathbf{\Sigma}_1^{-1} \mathbf{U}_1^H \mathbf{U}_1 \mathbf{\Sigma}_1 \mathbf{V}_1^H = \mathbf{V}_1 \mathbf{V}_1^H = \mathbf{I}$$

$L \times N \quad N \times L \quad L \times L$

- If  $\mathbf{A}$  is not full rank, i.e.  $r < L$ :

$$\underset{N \times L}{\mathbf{A}} = \underset{N \times N}{\mathbf{U}} \underset{N \times L}{\boldsymbol{\Sigma}} \underset{L \times L}{\mathbf{V}^H} = \underset{N \times r}{\mathbf{U}_1} \underset{r \times r}{\boldsymbol{\Sigma}_1} \underset{r \times L}{\mathbf{V}_1^H}$$

$$\underset{L \times N}{\mathbf{A}^\#} = \left( \underset{N \times N}{\mathbf{U}} \underset{N \times L}{\boldsymbol{\Sigma}} \underset{L \times L}{\mathbf{V}^H} \right)^\# = \left( \underset{L \times L}{\mathbf{V}^H} \right)^\# \boldsymbol{\Sigma}^\# \mathbf{U}^\# = \underset{L \times r}{\mathbf{V}} \boldsymbol{\Sigma}^\# \underset{r \times N}{\mathbf{U}^H} = \underset{L \times r}{\mathbf{V}_1} \boldsymbol{\Sigma}_1^{-1} \underset{r \times N}{\mathbf{U}_1^H}$$

where:

$$\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_1 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \boldsymbol{\Sigma}^\# \triangleq \begin{bmatrix} \boldsymbol{\Sigma}_1^{-1} & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \boldsymbol{\Sigma}^\# \boldsymbol{\Sigma} = \begin{bmatrix} \mathbf{I} & 0 \\ 0 & 0 \end{bmatrix}$$

$$\underset{L \times N}{\mathbf{A}^\#} \underset{N \times L}{\mathbf{A}} = \underset{L \times r}{\mathbf{V}_1} \boldsymbol{\Sigma}_1^{-1} \underset{r \times N}{\mathbf{U}_1^H} \underset{N \times L}{\mathbf{U}_1} \boldsymbol{\Sigma}_1 \underset{L \times L}{\mathbf{V}_1^H} = \underset{L \times L}{\mathbf{V}_1} \underset{r \times r}{\mathbf{V}_1^H} \neq \underset{L \times L}{\mathbf{I}} \quad \left( \underset{r \times r}{\mathbf{V}_1^H \mathbf{V}_1} = \underset{r \times r}{\mathbf{I}} \right)$$

- If  $\mathbf{A}$  is a square full-rank matrix:  $\mathbf{A}^\# = \mathbf{A}^{-1}$ , if it is also Hermitian:  $\mathbf{V} = \mathbf{U}$

- If the square matrix  $\mathbf{A}$  is not Hermitian and some eigenvalues have geometric multiplicity (i.e. the number of linearly independent eigenvectors related to that eigenvalue) lower than its algebraic multiplicity ( $>1$ ), the eigenvectors are linearly dependent, so the matrix  $\mathbf{U}$  of the eigenvectors cannot be inverted and matrix  $\mathbf{A}$  cannot be factorized (diagonalized), as previously described.

$$\mathbf{A}\mathbf{U} = \mathbf{U}\mathbf{\Sigma} \xRightarrow{\mathbf{U} \text{ full-rank}} \mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{U}^{-1} \xRightarrow{\mathbf{A} \text{ Hermitian}} \mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{U}^H$$

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad EVD: \quad \mathbf{A}\mathbf{U} = \mathbf{U}\mathbf{\Sigma} \Rightarrow \mathbf{U} = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{\Sigma} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\mathbf{A}^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

- $\mathbf{A}$  cannot be factorized as  $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{U}^{-1}$  since matrix  $\mathbf{U}$  cannot be inverted, so we cannot derive the inverse of  $\mathbf{A}$  as  $\mathbf{A}^{-1} = \mathbf{U}\mathbf{\Sigma}^{-1}\mathbf{U}^{-1}$ .



- Instead, the SVD of the (square) matrix **A** exists and it is given by:

$$SVD: \mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H \Rightarrow \mathbf{U} = \begin{bmatrix} 0.8507 & -0.5257 \\ 0.5257 & 0.8507 \end{bmatrix},$$

$$\mathbf{\Sigma} = \begin{bmatrix} 1.618 & 0 \\ 0 & 0.618 \end{bmatrix}, \quad \mathbf{V} = \begin{bmatrix} 0.5257 & -0.8507 \\ 0.8507 & 0.5257 \end{bmatrix}.$$

$$\mathbf{A}^\# = \mathbf{V}\mathbf{\Sigma}^{-1}\mathbf{U}^H = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \mathbf{A}^{-1}$$



SVD of  $\mathbf{A}$  when  $\text{rank}(\mathbf{A})=r$ , with  $r \leq L < N$

$$\mathbf{A}_{N \times L} = \mathbf{U}_{N \times N} \mathbf{\Sigma}_{N \times L} \mathbf{V}_{L \times L}^H = [\mathbf{U}_1 \quad \mathbf{U}_2] \begin{bmatrix} \mathbf{\Sigma}_1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{V}_1^H \\ \mathbf{V}_2^H \end{bmatrix} = \mathbf{U}_1 \mathbf{\Sigma}_1 \mathbf{V}_1^H$$

$$\begin{matrix} N \times r & r \times r & r \times L \end{matrix}$$

$$\mathbf{A}^\#_{L \times N} = \mathbf{V}_1_{L \times r} \mathbf{\Sigma}_1^{-1}_{r \times r} \mathbf{U}_1^H_{r \times N}$$

■ Least Squares (LS) solution of the overdetermined linear system:

$$\mathbf{b} = \mathbf{A}\mathbf{x} + \mathbf{e}$$

**Problem:** find  $\mathbf{x}$  such that  $\|\mathbf{e}\|^2 = \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$  is minimum.

**Solution:**  $\hat{\mathbf{x}}_{LS} = \mathbf{A}^\# \mathbf{b}$ , where  $\mathbf{A}^\#_{L \times N} = (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H = \mathbf{V}_1 \mathbf{\Sigma}_1^{-1} \mathbf{U}_1^H$



# **The Deterministic Linear Model**

■ When the signal of interest (SOI)  $\mathbf{S}$  can be expressed as a linear combination of  $p$  vectors  $\mathbf{h}_i$ , i.e.  $\mathbf{S}$  belongs to the linear subspace generated by the  $p$  vectors  $\mathbf{h}_i$ , with  $p \leq N$ , then we have:

$$\mathbf{X}_{N \times 1} = \mathbf{S} + \mathbf{W} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W} = \sum_{i=1}^p \theta_i \mathbf{h}_i + \mathbf{W}$$

$$\mathbf{H}_{N \times p} \triangleq [\mathbf{h}_1 \quad \mathbf{h}_2 \quad \cdots \quad \mathbf{h}_p], \quad \boldsymbol{\theta}_{p \times 1} \triangleq \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_p \end{bmatrix}$$

$\mathbf{X}$  = observed data vector

$\mathbf{W}$  = measurement error and/or modelling error

$N$  = number of measurements (number of equations)

$p$  = number of unknowns

$N > p \rightarrow$  overdetermined system ( $\mathbf{H}$  is a "tall matrix")

■ **Problem:** Given the observed data vector  $\mathbf{X}$  and given  $\mathbf{H}$ , a priori known, derive an estimate of the deterministic unknown parameter vector  $\boldsymbol{\theta}$ .

■ **Linear Least Squares (LS) estimate:**

$$\hat{\boldsymbol{\theta}}_{LS} = \arg \min_{\boldsymbol{\theta}} \|\mathbf{X} - \mathbf{H}\boldsymbol{\theta}\|^2 \rightarrow \hat{\boldsymbol{\theta}}_{LS} = \mathbf{H}^{\#} \mathbf{X} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \mathbf{X}$$

■ Note that once  $\boldsymbol{\theta}$  has been estimated, we can estimate the signal as:

$$\hat{\mathbf{S}}_{LS} = \mathbf{H} \hat{\boldsymbol{\theta}}_{LS}$$

- Let us go back to our problem of estimating the parameters of a deterministic real-valued signal observed in the presence of AWGN, given  $N$  samples of the received signal:

$$\mathbf{X} = \mathbf{s}(\boldsymbol{\theta}) + \mathbf{W}, \quad \mathbf{W} \in \mathcal{N}(\mathbf{0}, \mathbf{R}), \quad \mathbf{R} = E\{\mathbf{W}\mathbf{W}^T\} = \sigma_w^2 \mathbf{I}$$

- **Problem:** Assuming that the relationship between the signal and the unknown parameters is known, derive the **Maximum Likelihood (ML)** estimator of  $\boldsymbol{\theta}$ .
- **Assumption:** the real-valued noise is additive, white, and Gaussian (AWGN), so the joint pdf is:

$$f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) = \frac{1}{\sqrt{(2\pi)^N |\mathbf{R}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{s}(\boldsymbol{\theta}))^T \mathbf{R}^{-1}(\mathbf{x} - \mathbf{s}(\boldsymbol{\theta}))\right)$$

$$|\mathbf{R}| = \det\{\mathbf{R}\}, \quad \mathbf{R} = \mathbf{R}^T \quad \text{positive definite (so symmetric) matrix}$$



$$\mathbf{R} = \sigma_w^2 \mathbf{I} \Rightarrow f_{\mathbf{x}}(\mathbf{x}; \boldsymbol{\theta}) = \frac{1}{\sqrt{(2\pi\sigma_w^2)^N}} \exp\left(-\frac{1}{2\sigma_w^2} (\mathbf{x} - \mathbf{s}(\boldsymbol{\theta}))^T (\mathbf{x} - \mathbf{s}(\boldsymbol{\theta}))\right)$$

$$\ln L(\boldsymbol{\theta}) = \ln f_{\mathbf{x}}(\mathbf{x}; \boldsymbol{\theta}) = \kappa - \frac{1}{2\sigma_w^2} (\mathbf{x} - \mathbf{s}(\boldsymbol{\theta}))^T (\mathbf{x} - \mathbf{s}(\boldsymbol{\theta}))$$

$$= \kappa - \frac{1}{2\sigma_w^2} \sum_{n=0}^{N-1} (x_n - s_n(\boldsymbol{\theta}))^2$$

$$= \kappa - \frac{1}{2\sigma_w^2} \|\mathbf{x} - \mathbf{s}(\boldsymbol{\theta})\|^2$$



$$\hat{\boldsymbol{\theta}}_{ML} = \arg \max_{\boldsymbol{\theta}} \ln L(\boldsymbol{\theta}) = \arg \min_{\boldsymbol{\theta}} \|\mathbf{x} - \mathbf{s}(\boldsymbol{\theta})\|^2 = \hat{\boldsymbol{\theta}}_{LS}$$



$$AWGN \Rightarrow \hat{\boldsymbol{\theta}}_{ML} = \arg \max_{\boldsymbol{\theta}} \ln f_{\mathbf{x}}(\mathbf{x}; \boldsymbol{\theta}) = \arg \min_{\boldsymbol{\theta}} \|\mathbf{x} - \mathbf{s}(\boldsymbol{\theta})\|^2 = \hat{\boldsymbol{\theta}}_{LS}$$

■ Hence, if the measurement noise  $\mathbf{w}$  is AWGN, the ML estimate and the LS estimate coincide.

■ However, the LS can be applied even when the noise is not white and/or non-Gaussian, i.e. even when we do not have a statistical model of the measurement error  $\mathbf{w}$ .

■ **ML estimation** requires knowledge of the data pdf, i.e. knowledge of the parametric signal model  $\mathbf{s}(\boldsymbol{\theta})$  and the statistical model of the measurement noise  $\mathbf{w}$ .

■ **LS estimation** requires only knowledge of the parametric signal model  $\mathbf{s}(\boldsymbol{\theta})$ .

- **Assumption:** the signal can be modelled by the Deterministic Linear Model.

$$\mathbf{s}(\boldsymbol{\theta}) = \mathbf{H}\boldsymbol{\theta}$$

$$\mathbf{X} \in \mathcal{N}(\mathbf{H}\boldsymbol{\theta}, \mathbf{R}) \Leftrightarrow \begin{cases} \boldsymbol{\eta} \triangleq E\{\mathbf{X}\} = E\{\mathbf{H}\boldsymbol{\theta} + \mathbf{W}\} = \mathbf{H}\boldsymbol{\theta} \\ \mathbf{R} = E\{(\mathbf{X} - \boldsymbol{\eta})(\mathbf{X} - \boldsymbol{\eta})^T\} = E\{\mathbf{W}\mathbf{W}^T\} = \sigma_w^2 \mathbf{I} \quad (\text{AWGN}) \end{cases}$$

- The **log-Likelihood Function (log-LF)** is given by:

$$\begin{aligned} \ln L(\boldsymbol{\theta}) &= \ln f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) = \kappa - \frac{1}{2} (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T \mathbf{R}^{-1} (\mathbf{x} - \mathbf{H}\boldsymbol{\theta}) \\ &= \kappa - \frac{1}{2\sigma_w^2} (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T (\mathbf{x} - \mathbf{H}\boldsymbol{\theta}) = \kappa - \frac{1}{2\sigma_w^2} \|\mathbf{x} - \mathbf{H}\boldsymbol{\theta}\|^2 \end{aligned}$$



- The **ML estimate** of  $\theta$  can be obtained by calculating the gradient of  $\ln L(\theta)$ :

$$\hat{\theta}_{ML} = \arg \max_{\theta} \ln L(\theta) = \arg \max_{\theta} \left[ \kappa - \frac{1}{2\sigma_w^2} (\mathbf{x} - \mathbf{H}\theta)^T (\mathbf{x} - \mathbf{H}\theta) \right]$$

$$= \arg \min_{\theta} (\mathbf{x} - \mathbf{H}\theta)^T (\mathbf{x} - \mathbf{H}\theta) = \arg \min_{\theta} \|\mathbf{x} - \mathbf{H}\theta\|^2 = \hat{\theta}_{LS}$$

$$\nabla_{\theta} \ln L(\theta) = \frac{\partial \ln L(\theta)}{\partial \theta} = -\frac{1}{2\sigma_w^2} \frac{\partial \{(\mathbf{x} - \mathbf{H}\theta)^T (\mathbf{x} - \mathbf{H}\theta)\}}{\partial \theta}$$

$$= \frac{1}{\sigma_w^2} \left[ \frac{\partial (\mathbf{H}\theta)^T}{\partial \theta} (\mathbf{x} - \mathbf{H}\theta) \right]$$

$$= \frac{1}{\sigma_w^2} [\mathbf{H}^T (\mathbf{x} - \mathbf{H}\theta)] = 0$$

$$\Rightarrow \mathbf{H}^T \mathbf{x} = \mathbf{H}^T \mathbf{H} \theta \Rightarrow \theta = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x} = \mathbf{H}^{\#} \mathbf{x}$$

■ When the SOI is embedded in AWGN, the **ML estimate** coincides with the **LS estimate**. If the signal  $\mathbf{s}$  depends linearly on  $\boldsymbol{\theta}$ , the estimate is **linear** and **Gaussian** distributed (since  $\mathbf{X}$  is a Gaussian vector):

$$\hat{\boldsymbol{\theta}}_{ML} = \arg \min_{\boldsymbol{\theta}} J(\boldsymbol{\theta}) = \arg \min_{\boldsymbol{\theta}} \|\mathbf{x} - \mathbf{H}\boldsymbol{\theta}\|^2 = \hat{\boldsymbol{\theta}}_{LS}$$

where  $\hat{\boldsymbol{\theta}}_{LS} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x} = \mathbf{H}^{\#} \mathbf{x}$

$\mathbf{H}^{\#} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T$  Moore-Penrose pseudoinverse:  $\mathbf{H}^{\#}_{p \times N} \mathbf{H}_{N \times p} = \mathbf{I}_{p \times p}$

$E\{\hat{\boldsymbol{\theta}}_{ML}\} = E\{\mathbf{H}^{\#} \mathbf{X}\} = \mathbf{H}^{\#} E\{\mathbf{X}\} = \mathbf{H}^{\#} \mathbf{H} \boldsymbol{\theta} = \boldsymbol{\theta}$



■ The MSE matrix of the **ML estimate** is given by:

$$\begin{aligned} \text{MSE}\{\hat{\boldsymbol{\theta}}_{ML}\} &= E\{(\hat{\boldsymbol{\theta}}_{ML} - \boldsymbol{\theta})(\hat{\boldsymbol{\theta}}_{ML} - \boldsymbol{\theta})^T\} = E\{(\mathbf{H}^\# \mathbf{X} - \boldsymbol{\theta})(\mathbf{H}^\# \mathbf{X} - \boldsymbol{\theta})^T\} \\ &= E\{(\mathbf{H}^\# \mathbf{H} \boldsymbol{\theta} + \mathbf{H}^\# \mathbf{W} - \boldsymbol{\theta})(\mathbf{H}^\# \mathbf{H} \boldsymbol{\theta} + \mathbf{H}^\# \mathbf{W} - \boldsymbol{\theta})^T\} \\ &= E\{(\mathbf{H}^\# \mathbf{W})(\mathbf{H}^\# \mathbf{W})^T\} = \mathbf{H}^\# E\{\mathbf{W} \mathbf{W}^T\} (\mathbf{H}^\#)^T \\ &= \sigma_w^2 (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \left[ (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \right]^T \\ &= \sigma_w^2 (\mathbf{H}^T \mathbf{H})^{-1} \end{aligned}$$

■ The estimator is **unbiased** and **efficient**, in addition to be **Gaussian** and **consistent**:

$$\begin{aligned}\frac{\partial \ln L(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} &= \frac{1}{\sigma_w^2} [\mathbf{H}^T (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})] = \frac{1}{\sigma_w^2} [\mathbf{H}^T \mathbf{x} - \mathbf{H}^T \mathbf{H} \boldsymbol{\theta}] \\ &= \frac{(\mathbf{H}^T \mathbf{H})}{\sigma_w^2} [(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x} - \boldsymbol{\theta}] = \mathbf{I}(\boldsymbol{\theta}) [\mathbf{g}(\mathbf{x}) - \boldsymbol{\theta}] \\ \Rightarrow &\begin{cases} FIM : \mathbf{I}(\boldsymbol{\theta}) = \frac{(\mathbf{H}^T \mathbf{H})}{\sigma_w^2} \rightarrow CRB(\boldsymbol{\theta}) = \sigma_w^2 (\mathbf{H}^T \mathbf{H})^{-1} \\ \hat{\boldsymbol{\theta}}_{ML} = \mathbf{g}(\mathbf{x}) = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x} = \mathbf{H}^\# \mathbf{x} \end{cases}\end{aligned}$$


$$\hat{\boldsymbol{\theta}}_{ML} \in \mathcal{N}(\boldsymbol{\theta}, CRB(\boldsymbol{\theta}))$$

$$\mathbf{x} = \mathbf{s} + \mathbf{w} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$$

$$\hat{\boldsymbol{\theta}}_{LS} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x} = \mathbf{H}^\# \mathbf{x}$$

$$\hat{\mathbf{s}}_{LS} = \mathbf{H} \hat{\boldsymbol{\theta}}_{LS} = \mathbf{H} \mathbf{H}^\# \mathbf{x} = \mathbf{P}_H \mathbf{x}$$

$$\begin{aligned} \mathbf{e}_{\min} &\triangleq \mathbf{x} - \hat{\mathbf{s}}_{LS} = \mathbf{x} - \mathbf{H} \hat{\boldsymbol{\theta}}_{LS} = \mathbf{x} - \mathbf{H} \mathbf{H}^\# \mathbf{x} \\ &= (\mathbf{I} - \mathbf{H} \mathbf{H}^\#) \mathbf{x} = (\mathbf{I} - \mathbf{P}_H) \mathbf{x} = \mathbf{P}_H^\perp \mathbf{x} \end{aligned}$$

$$\begin{aligned} \mathbf{H}^T \mathbf{e}_{\min} &= \mathbf{H}^T \left[ \mathbf{x} - \mathbf{H} (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x} \right] \\ &= \mathbf{H}^T \mathbf{x} - \mathbf{H}^T \mathbf{H} (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x} \\ &= \mathbf{H}^T \mathbf{x} - \mathbf{H}^T \mathbf{x} = \mathbf{0} \\ &\rightarrow \mathbf{h}_k^T \mathbf{e} = \mathbf{0} \quad \forall k \end{aligned}$$

$$\mathbf{H}^T \mathbf{e}_{\min} = \mathbf{0} \Rightarrow \mathbf{h}_k^T \mathbf{e}_{\min} = 0 \quad \forall k$$

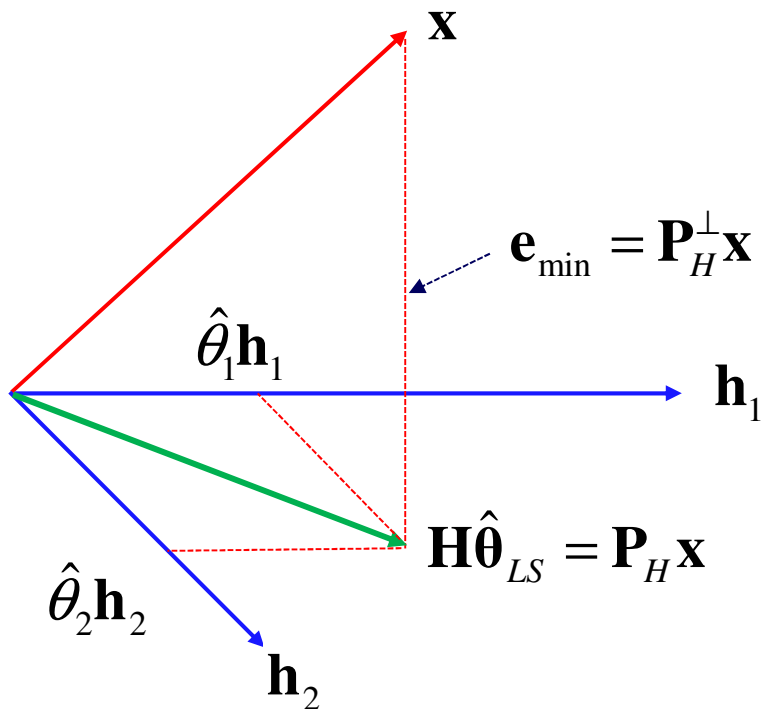
■ The signal LS estimate is the orthogonal projection of  $\mathbf{x}$  onto the subspace generated by the columns of matrix  $\mathbf{H}$ .

$$\mathbf{x} = \hat{\mathbf{s}}_{LS} + \mathbf{e}_{\min}$$

■ The error vector  $\mathbf{e}_{\min}$  is orthogonal to the subspace generated by the columns of matrix  $\mathbf{H}$ , i.e. it is orthogonal to all columns  $\mathbf{h}_k$ .

# Geometrical Interpretation of the Linear LS Estimator 46

- This result means that the error vector  $\mathbf{e}$  that has the minimum energy (which corresponds to the LS estimate) is orthogonal to the subspace generated by the columns of matrix  $\mathbf{H}$ , so the LS estimate of signal vector  $\mathbf{s}$  is given by the orthogonal projection of the observed signal vector  $\mathbf{x}$  onto the subspace generated by the columns of matrix  $\mathbf{H}$ . Hence, the error  $\mathbf{e}_{\min}$  is orthogonal to the estimate.
- This result is known as the **Orthogonality Principle** or **Projection Theorem**.



- Geometrical viewpoint of linear Least Squares in  $R^3$

$$\hat{\mathbf{s}}_{LS} = \mathbf{H}\hat{\boldsymbol{\theta}}_{LS} = \mathbf{P}_H \mathbf{x}$$

$$\mathbf{e}_{\min} = \mathbf{x} - \hat{\mathbf{s}}_{LS} = \mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}_{LS} = \mathbf{P}_H^\perp \mathbf{x}$$

$$\mathbf{e}_{\min} \perp \mathbf{h}_k \quad \forall k \quad i.e. \quad \mathbf{e}_{\min} \perp R(\mathbf{H})$$

where  $R(\mathbf{H})$  is the 2D subspace spanned by  $\mathbf{h}_1$  and  $\mathbf{h}_2$

- **Example:** we want to estimate the amplitude of a signal of known shape. In this case, the relationship between the signal of interest (SOI) and the unknown parameter is linear:

$$\mathbf{x} = \mathbf{s}(\boldsymbol{\theta}) + \mathbf{w} = A\mathbf{p} + \mathbf{w}$$

$$\mathbf{s}(\boldsymbol{\theta}) = \mathbf{H}\boldsymbol{\theta} = A\mathbf{p}, \text{ where } \boldsymbol{\theta} = A \text{ and } \mathbf{H}=\mathbf{p} \text{ is perfectly known}$$

$$\hat{A}_{LS} = \arg \min_A \|\mathbf{x} - A\mathbf{p}\|^2$$

- It is important to stress that the **LS method** can be applied even when we do not have information about the statistical model of the noise, we just need to know the signal model  $\mathbf{s}(\boldsymbol{\theta})$ .
- The **LS** method → is a purely deterministic approach.

$$\begin{aligned}\hat{A}_{LS} &= \arg \min_A \|\mathbf{x} - A\mathbf{p}\|^2 \\ &= \mathbf{p}^\# \mathbf{x} = (\mathbf{p}^H \mathbf{p})^{-1} \mathbf{p}^H \mathbf{x} \\ &= \frac{\mathbf{p}^H \mathbf{x}}{\mathbf{p}^H \mathbf{p}} = \frac{\mathbf{p}^H \mathbf{x}}{\|\mathbf{p}\|^2} \\ &= \frac{1}{E_p} \mathbf{p}^H \mathbf{x}, \quad \text{where } E_p \triangleq \|\mathbf{p}\|^2 \text{ is the energy of vector } \mathbf{p}\end{aligned}$$

$$\hat{A}_{LS} = \frac{1}{E_p} \mathbf{p}^H \mathbf{x} = \frac{1}{E_p} \sum_{n=0}^{N-1} p_n^* x_n \quad \rightarrow \quad \text{FIR Matched Filter (matched to } \mathbf{p}\text{)}$$

$$p_n = 1 \quad \forall n \quad \rightarrow \quad E_p = N \quad \rightarrow \quad \hat{A}_{LS} = \frac{1}{N} \sum_{n=0}^{N-1} x_n \quad \rightarrow \quad \text{Sample Mean}$$



■ **Example:** we want to reveal the presence of a **linear trend** in the observed data.

$$X[n] = A + Bn + W[n], \quad 0 \leq n \leq N-1 \quad \text{Ex: Dow Jones}$$

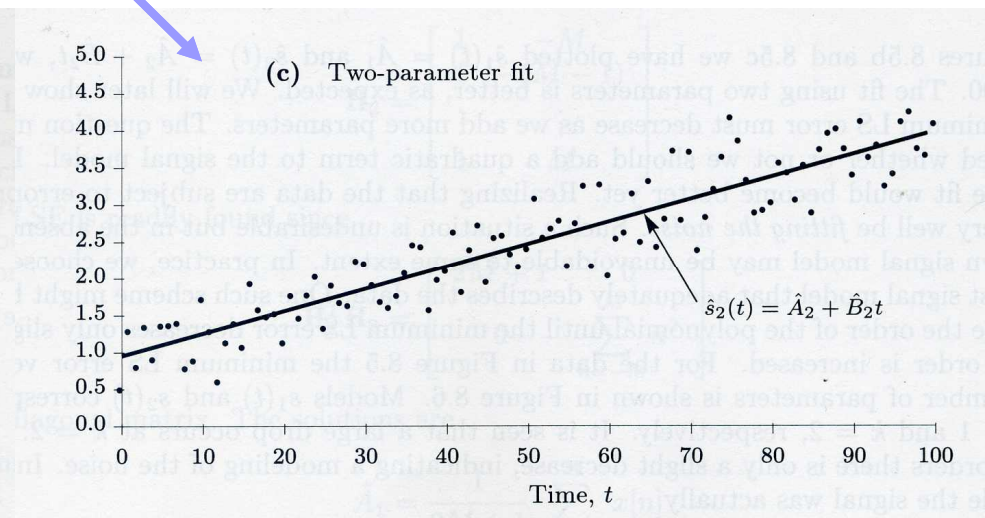
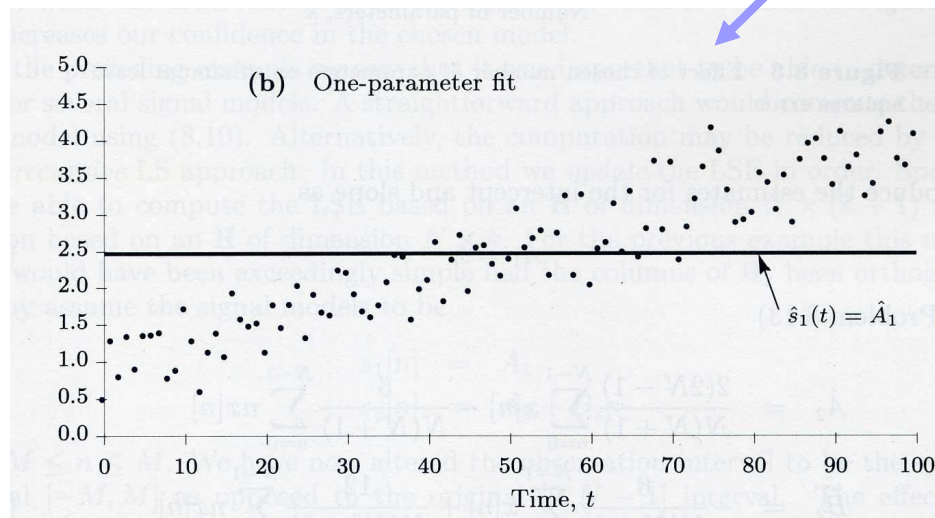
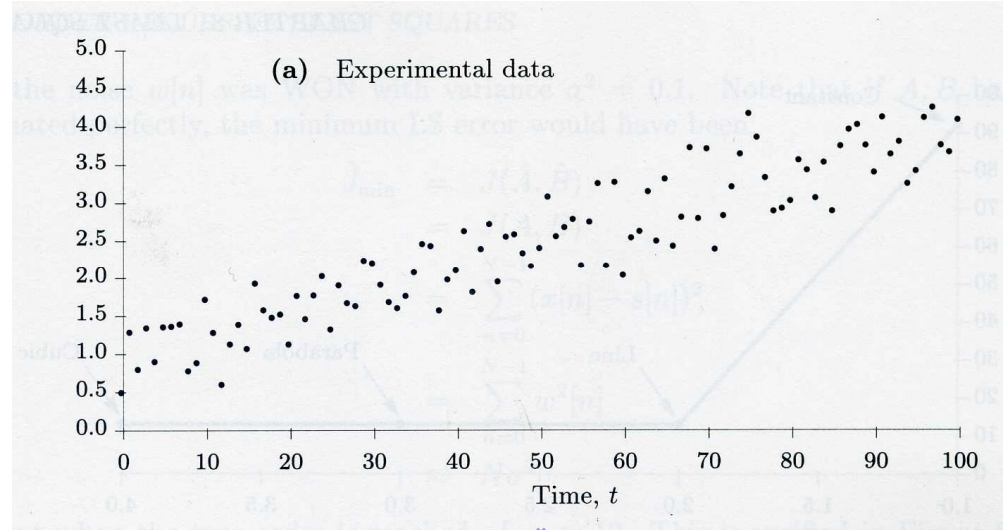
$$\mathbf{X}_{N \times 1} = \begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & N-1 \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} + \mathbf{W} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W}$$

$$\text{where } \mathbf{H}_{N \times 2} \triangleq \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & N-1 \end{bmatrix} \quad \text{and} \quad \boldsymbol{\theta}_{2 \times 1} \triangleq \begin{bmatrix} A \\ B \end{bmatrix}$$

$$\hat{\boldsymbol{\theta}}_{LS} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \mathbf{X} = \mathbf{H}^\# \mathbf{X}$$

# The Linear Model

50



50



$$\begin{aligned}\mathbf{H}^\# &= \left( \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 0 & 1 & \cdots & N-1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & N-1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 0 & 1 & \cdots & N-1 \end{bmatrix} \\ &= \begin{bmatrix} N & \sum_{k=0}^{N-1} k \\ \sum_{k=0}^{N-1} k & \sum_{k=0}^{N-1} k^2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 0 & 1 & \cdots & N-1 \end{bmatrix} \\ &= \begin{bmatrix} N & \frac{N(N-1)}{2} \\ \frac{N(N-1)}{2} & \frac{N(N-1)(2N-1)}{6} \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 0 & 1 & \cdots & N-1 \end{bmatrix}\end{aligned}$$



$$\begin{aligned}\mathbf{H}^\# &= \frac{2}{N} \begin{bmatrix} 2 & N-1 \\ N-1 & \frac{(N-1)(2N-1)}{3} \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 & \dots & 1 \\ 0 & 1 & \dots & N-1 \end{bmatrix} \\ &= \frac{6}{N(N-1)[2(2N-1)-3(N-1)]} \begin{bmatrix} \frac{(N-1)(2N-1)}{3} & -(N-1) \\ -(N-1) & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & \dots & 1 \\ 0 & 1 & \dots & N-1 \end{bmatrix}\end{aligned}$$

$$\hat{\boldsymbol{\theta}}_{LS} = \mathbf{H}^\# \mathbf{X} = \frac{6}{N(N-1)(N+1)} \begin{bmatrix} \frac{(N-1)(2N-1)}{3} & -(N-1) \\ -(N-1) & 2 \end{bmatrix} \begin{bmatrix} \sum_{n=0}^{N-1} X[n] \\ \sum_{n=0}^{N-1} n \cdot X[n] \end{bmatrix}$$

$$\hat{A}_{LS} = \frac{2(2N-1)}{N(N+1)} \sum_{n=0}^{N-1} X[n] - \frac{6}{N(N+1)} \sum_{n=0}^{N-1} n \cdot X[n]$$

$$\hat{B}_{LS} = -\frac{6}{N(N+1)} \sum_{n=0}^{N-1} X[n] + \frac{12}{N(N^2-1)} \sum_{n=0}^{N-1} n \cdot X[n]$$

- If the noise is AWGN, the LS estimators are also ML estimators.
- As we already proved, the ML estimator of the Deterministic Linear model in AWGN is efficient, therefore:

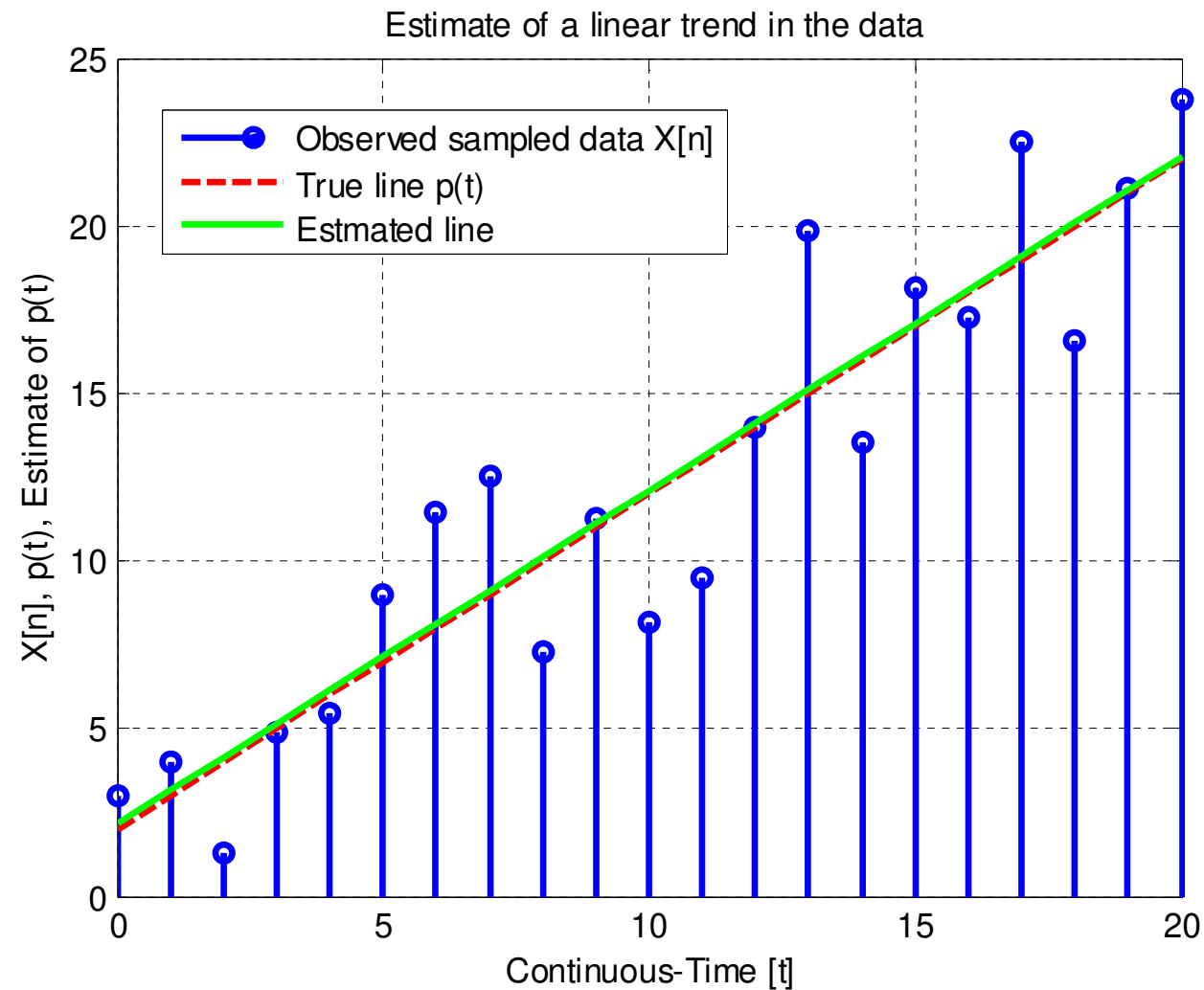
$$MSE\{\hat{A}_{ML}\} = CRB(A) = \sigma_w^2 \left[ (\mathbf{H}^H \mathbf{H})^{-1} \right]_{1,1} = \frac{2(2N-1)}{N(N+1)} \sigma_w^2 \propto \frac{1}{N}$$

$$MSE\{\hat{B}_{ML}\} = CRB(B) = \sigma_w^2 \left[ (\mathbf{H}^H \mathbf{H})^{-1} \right]_{2,2} = \frac{12}{N(N^2-1)} \sigma_w^2 \propto \frac{1}{N^3}$$

- This result suggests that we can estimate more accurately the slope  $B$  than the parameter  $A$ , since the observed data are more strongly affected by  $B$ .

$$p(t) = A + Bt, \quad X(t) = p(t) + W(t), \quad t_k = kT_c, \quad \{W(t_k)\}_{k=0}^{N-1} \in \mathcal{N}(0, \sigma_W^2)$$

$$\begin{aligned} T_c &= 1 \\ N &= 21 \\ A &= 2 \\ B &= 1 \\ \sigma_W^2 &= 10 \end{aligned}$$

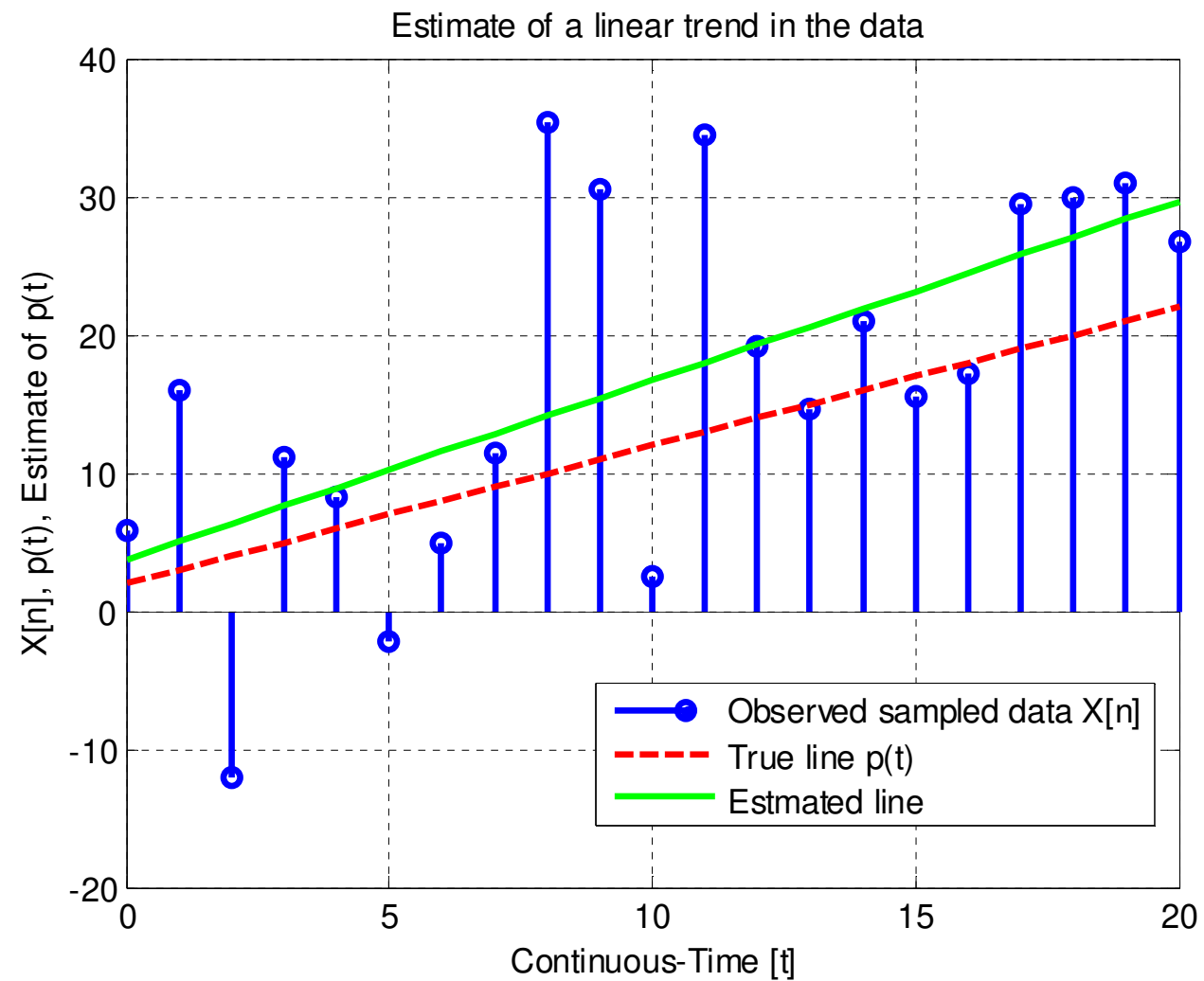




# The Linear Model

55

$$\begin{aligned}T_c &= 1 \\N &= 21 \\A &= 2 \\B &= 1 \\\sigma_w^2 &= 50\end{aligned}$$



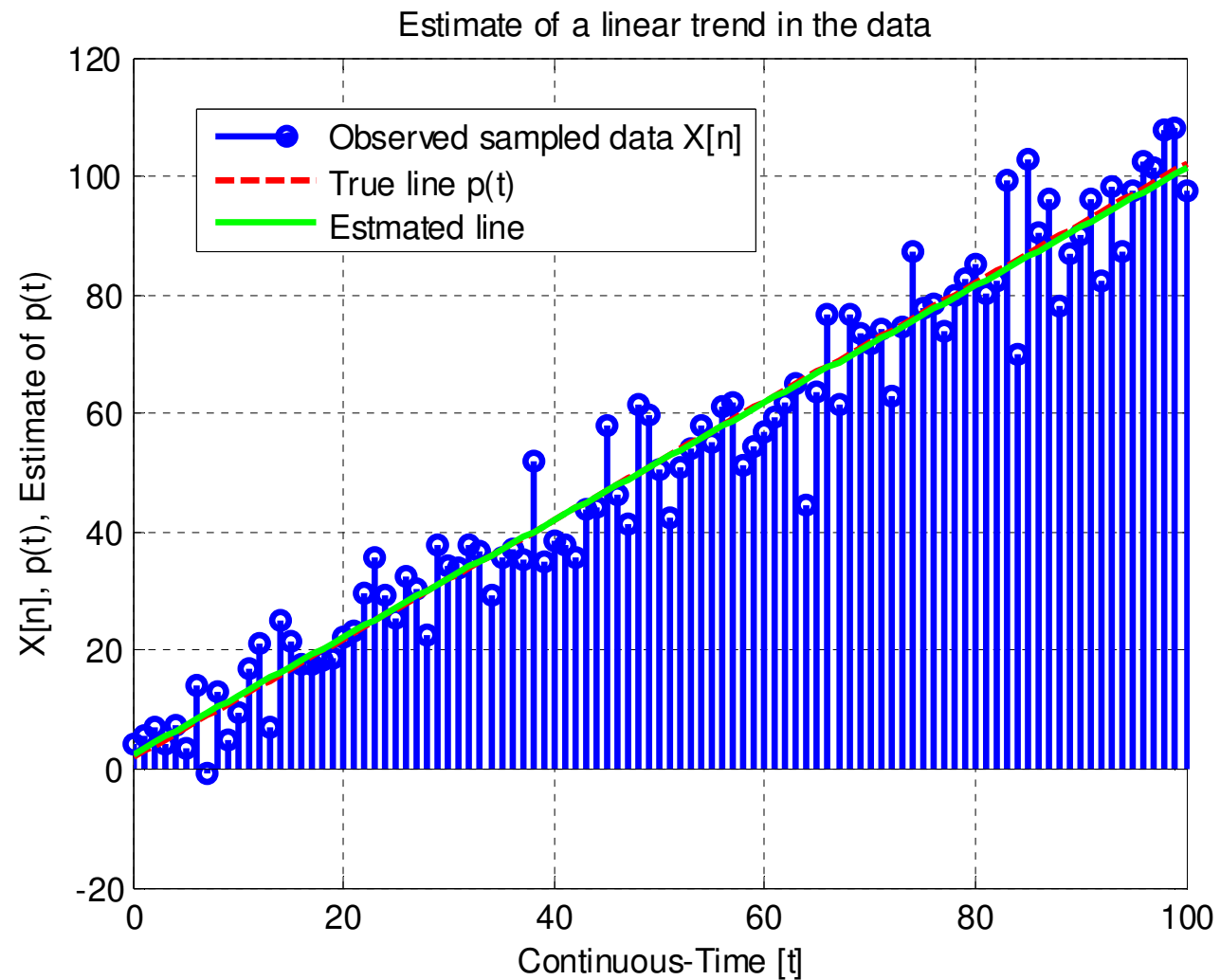
$$T_c = 1$$

$$N = 101$$

$$A = 2$$

$$B = 1$$

$$\sigma_w^2 = 50$$





■ **Example:** we want to estimate the amplitude and phase of a sinusoid in AWGN.

$$\begin{aligned} X[n] &= \frac{A}{\sqrt{2B}} \cos(2\pi F_0 n + \theta) + W[n] \\ &= \frac{A}{\sqrt{2B}} \cos(\theta) \cos(2\pi F_0 n) - \frac{A}{\sqrt{2B}} \sin(\theta) \sin(2\pi F_0 n) + W[n], \quad 0 \leq n \leq N-1 \end{aligned}$$

$$\begin{cases} \theta_1 \triangleq \frac{A}{\sqrt{2B}} \cos(\theta) \\ \theta_2 \triangleq -\frac{A}{\sqrt{2B}} \sin(\theta) \end{cases} \quad \begin{cases} \mathbf{h}_1 \triangleq [1 \quad \cos(2\pi F_0) \quad \cdots \quad \cos(2\pi F_0(N-1))]^T \\ \mathbf{h}_2 \triangleq [0 \quad \sin(2\pi F_0) \quad \cdots \quad \sin(2\pi F_0(N-1))]^T \end{cases}$$

$$\mathbf{X} = \theta_1 \mathbf{h}_1 + \theta_2 \mathbf{h}_2 + \mathbf{W} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W}$$

$$F_0 \triangleq \frac{f_0}{2B} \quad \text{digital frequency}$$

- This is the so-called “Deterministic Linear Model”:

$$\mathbf{X} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W}, \quad \text{where } \boldsymbol{\theta} \triangleq \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \quad \mathbf{H} \triangleq [\mathbf{h}_1 \quad \mathbf{h}_2]$$

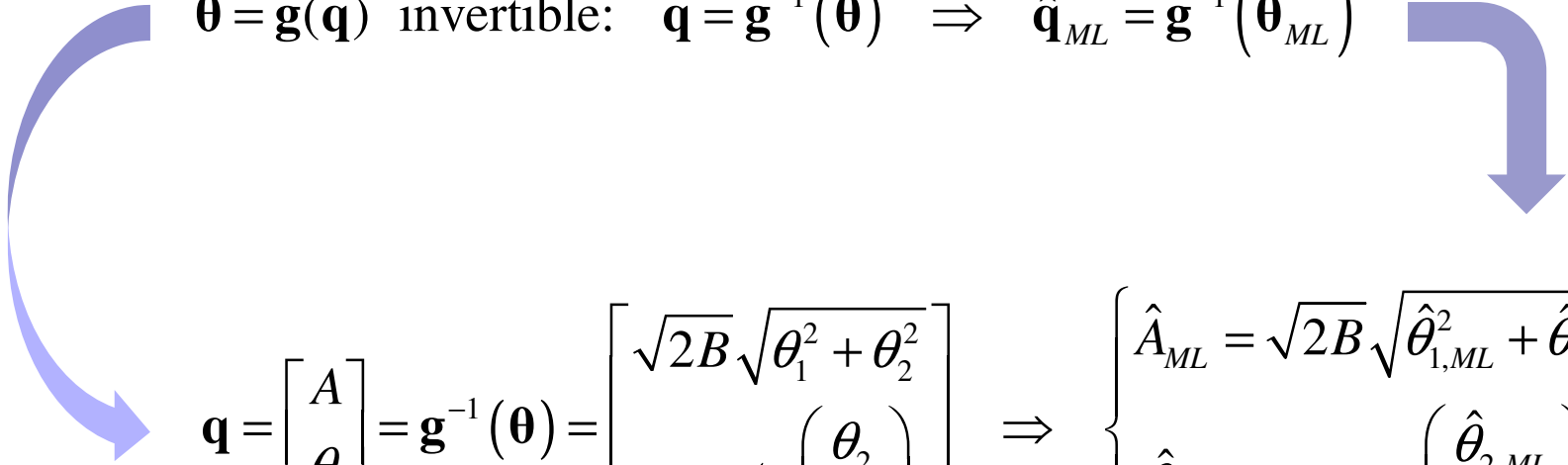
- where:  $\boldsymbol{\theta} \triangleq \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = \mathbf{g}(\mathbf{q}) = \begin{bmatrix} \frac{A}{\sqrt{2B}} \cos(\theta) \\ -\frac{A}{\sqrt{2B}} \sin(\theta) \end{bmatrix}, \quad \text{and } \mathbf{q} \triangleq \begin{bmatrix} A \\ \theta \end{bmatrix}$

- **Problem:** estimate  $\theta$ , given the observed data vector  $\mathbf{X}$ , assuming that  $\mathbf{H}$  is *a-priori* known.



■ Once we have derived the ML estimate of vector  $\theta$ , the ML estimate of the vector  $\mathbf{q}$  can be immediately derived thanks to the **invariance property** of the ML estimate:

$$\theta = \mathbf{g}(\mathbf{q}) \text{ invertible: } \mathbf{q} = \mathbf{g}^{-1}(\theta) \Rightarrow \hat{\mathbf{q}}_{ML} = \mathbf{g}^{-1}(\hat{\theta}_{ML})$$


$$\mathbf{q} = \begin{bmatrix} A \\ \theta \end{bmatrix} = \mathbf{g}^{-1}(\theta) = \begin{bmatrix} \sqrt{2B} \sqrt{\theta_1^2 + \theta_2^2} \\ -\arctg\left(\frac{\theta_2}{\theta_1}\right) \end{bmatrix} \Rightarrow \begin{cases} \hat{A}_{ML} = \sqrt{2B} \sqrt{\hat{\theta}_{1,ML}^2 + \hat{\theta}_{2,ML}^2} \\ \hat{\theta}_{ML} = -\arctg\left(\frac{\hat{\theta}_{2,ML}}{\hat{\theta}_{1,ML}}\right) \end{cases}$$

- It is worth observing that in the case we want to estimate the parameters of a sinusoid of unknown amplitude, phase, and **frequency**, matrix **H** is not perfectly known, but it is a known function of the unknown parameter  $F_0$ , i.e. it is parametrized by the unknown parameter  $F_0$ :

$$\mathbf{X}_{N \times 1} = \mathbf{H}(F_0)\boldsymbol{\theta} + \mathbf{W} = \sum_{i=1}^p \theta_i \mathbf{h}_i(F_0) + \mathbf{W}$$

$$\mathbf{H}(F_0)_{N \times 2} = [\mathbf{h}_1(F_0) \quad \mathbf{h}_2(F_0)] = \begin{bmatrix} 1 & \cos(2\pi F_0) & \cdots & \cos(2\pi F_0(N-1)) \\ 0 & \sin(2\pi F_0) & \cdots & \sin(2\pi F_0(N-1)) \end{bmatrix}^T$$

■ Assume first that  $F_0$  is known and we want to estimate **amplitude** and **phase**:

$$\hat{\boldsymbol{\theta}}_{ML} = \begin{bmatrix} \hat{\theta}_{1,ML} \\ \hat{\theta}_{2,ML} \end{bmatrix} = \mathbf{H}^\# \mathbf{x} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x} \Rightarrow \begin{cases} \hat{A}_{ML} = \sqrt{2B} \sqrt{\hat{\theta}_{1,ML}^2 + \hat{\theta}_{2,ML}^2} \\ \hat{\theta}_{ML} = -\arctg\left(\frac{\hat{\theta}_{2,ML}}{\hat{\theta}_{1,ML}}\right) \end{cases}$$

$$\mathbf{H} = [\mathbf{h}_1 \quad \mathbf{h}_2], \quad \text{where: } \begin{cases} \mathbf{h}_1 \triangleq [1 \quad \cos(2\pi F_0) \quad \cdots \quad \cos(2\pi F_0(N-1))]^T \\ \mathbf{h}_2 \triangleq [0 \quad \sin(2\pi F_0) \quad \cdots \quad \sin(2\pi F_0(N-1))]^T \end{cases}$$

$$\begin{aligned} \mathbf{H}^\# &= (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T = \left( \begin{bmatrix} \mathbf{h}_1^T \\ \mathbf{h}_2^T \end{bmatrix} [\mathbf{h}_1 \quad \mathbf{h}_2] \right)^{-1} \begin{bmatrix} \mathbf{h}_1^T \\ \mathbf{h}_2^T \end{bmatrix} = \begin{bmatrix} \mathbf{h}_1^T \mathbf{h}_1 & \mathbf{h}_1^T \mathbf{h}_2 \\ \mathbf{h}_2^T \mathbf{h}_1 & \mathbf{h}_2^T \mathbf{h}_2 \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{h}_1^T \\ \mathbf{h}_2^T \end{bmatrix} \\ &= \begin{bmatrix} \frac{N}{2} & 0 \\ 0 & \frac{N}{2} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{h}_1^T \\ \mathbf{h}_2^T \end{bmatrix} = \frac{2}{N} \begin{bmatrix} \mathbf{h}_1^T \\ \mathbf{h}_2^T \end{bmatrix} \end{aligned}$$

■ In fact, if  $N \gg 1$  we have:

$$\begin{aligned}\mathbf{h}_1^T \mathbf{h}_1 &= \|\mathbf{h}_1\|^2 = \sum_{n=0}^{N-1} \cos^2(2\pi F_0 n) = \sum_{n=0}^{N-1} \left( \frac{1}{2} + \frac{1}{2} \cos(4\pi F_0 n) \right) \\ &= \frac{N}{2} \left( 1 + \frac{1}{N} \sum_{n=0}^{N-1} \cos(4\pi F_0 n) \right) \cong \frac{N}{2}\end{aligned}$$

$$\begin{aligned}\mathbf{h}_2^T \mathbf{h}_2 &= \|\mathbf{h}_2\|^2 = \sum_{n=0}^{N-1} \sin^2(2\pi F_0 n) = \sum_{n=0}^{N-1} \left( \frac{1}{2} - \frac{1}{2} \cos(4\pi F_0 n) \right) \\ &= \frac{N}{2} \left( 1 - \frac{1}{N} \sum_{n=0}^{N-1} \cos(4\pi F_0 n) \right) \cong \frac{N}{2}\end{aligned}$$

$$\mathbf{h}_1^T \mathbf{h}_2 = \mathbf{h}_2^T \mathbf{h}_1 = \sum_{n=0}^{N-1} \cos(2\pi F_0 n) \sin(2\pi F_0 n) = \frac{1}{2} \sum_{n=0}^{N-1} \cos(4\pi F_0 n) \cong 0$$

- The ML/Linear LS estimate of vector  $\theta$  and its MSE are finally derived as:

$$\hat{\theta}_{ML} = \begin{bmatrix} \hat{\theta}_{1,ML} \\ \hat{\theta}_{2,ML} \end{bmatrix} = \mathbf{H}^\# \mathbf{x} = \frac{2}{N} \begin{bmatrix} \mathbf{h}_1^T \\ \mathbf{h}_2^T \end{bmatrix} \mathbf{x} = \begin{bmatrix} \frac{2}{N} \mathbf{h}_1^T \mathbf{x} \\ \frac{2}{N} \mathbf{h}_2^T \mathbf{x} \end{bmatrix} = \begin{bmatrix} \frac{2}{N} \sum_{n=0}^{N-1} X[n] \cos(2\pi F_0 n) \\ \frac{2}{N} \sum_{n=0}^{N-1} X[n] \sin(2\pi F_0 n) \end{bmatrix}$$

$$MSE\{\hat{\theta}_{ML}\} = \sigma_w^2 (\mathbf{H}^T \mathbf{H})^{-1} = \sigma_w^2 \begin{bmatrix} \frac{N}{2} & 0 \\ 0 & \frac{N}{2} \end{bmatrix}^{-1} = \begin{bmatrix} \frac{2\sigma_w^2}{N} & 0 \\ 0 & \frac{2\sigma_w^2}{N} \end{bmatrix} = \frac{2\sigma_w^2}{N} \mathbf{I}$$

$$\begin{cases} \hat{A}_{ML} = \sqrt{2B} \sqrt{\hat{\theta}_{1,ML}^2 + \hat{\theta}_{2,ML}^2} \\ \hat{\theta}_{ML} = -\arctg\left(\frac{\hat{\theta}_{2,ML}}{\hat{\theta}_{1,ML}}\right) \end{cases}$$

- The estimator **consistent**

$$\hat{\boldsymbol{\theta}}_{ML} = \hat{\boldsymbol{\theta}}_{LS} = \mathbf{H}^{\#} \mathbf{x} = \begin{bmatrix} \frac{2}{N} \sum_{n=0}^{N-1} X[n] \cos(2\pi F_0 n) \\ \frac{2}{N} \sum_{n=0}^{N-1} X[n] \sin(2\pi F_0 n) \end{bmatrix}$$

$$\left\{ \begin{array}{l} \hat{A}_{ML} = \frac{2\sqrt{2B}}{N} \sqrt{\left( \sum_{n=0}^{N-1} X[n] \cos(2\pi F_0 n) \right)^2 + \left( \sum_{n=0}^{N-1} X[n] \sin(2\pi F_0 n) \right)^2} \\ \hat{\theta}_{ML} = -\arctg \left( \frac{\sum_{n=0}^{N-1} X[n] \sin(2\pi F_0 n)}{\sum_{n=0}^{N-1} X[n] \cos(2\pi F_0 n)} \right) \end{array} \right.$$

■ They coincide with the previously derived expressions (Ch. 3, p. 137).



- If the frequency  $F_0$  is unknown,  $\mathbf{H}$  is also unknown. To be more precise,  $\mathbf{H}$  is a known function of the unknown parameter  $F_0$ :

$$\left(\hat{\boldsymbol{\theta}}_{ML}, \hat{F}_{0,ML}\right) = \arg \min_{\boldsymbol{\theta}, F_0} J(\boldsymbol{\theta}, F_0) = \arg \min_{\boldsymbol{\theta}, F_0} \|\mathbf{x} - \mathbf{H}(F_0)\boldsymbol{\theta}\|^2$$

$$\hat{\boldsymbol{\theta}}_{ML}(F_0) = \hat{\boldsymbol{\theta}}_{LS}(F_0) = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x} = \mathbf{H}^\# \mathbf{x}$$

$$\mathbf{H}(F_0) = \begin{bmatrix} \mathbf{h}_1(F_0) & \mathbf{h}_2(F_0) \end{bmatrix} = \begin{bmatrix} 1 & \cos(2\pi F_0) & \cdots & \cos(2\pi F_0(N-1)) \\ 0 & \sin(2\pi F_0) & \cdots & \sin(2\pi F_0(N-1)) \end{bmatrix}^T$$

$N \times 2$

- The **ML estimate** of  $F_0$  coincides with the **Non-Linear Least Squares (NLLS)** estimate: it is the value that minimizes the energy of the error between the data  $\mathbf{x}$  and the model. It is nonlinear since the signal depends on  $F_0$  in a nonlinear fashion.

- Once we have derived the ML (i.e. the linear LS) estimate of  $\theta$ , we insert the estimate in the functional  $J(\cdot)$  to be minimized and derive the estimate of  $F_0$  by minimizing the “compressed” functional w.r.t.  $F_0$ :

$$\begin{aligned}\hat{F}_{0,ML} &= \hat{F}_{0,NLLS} = \arg \min_{F_0} J\left(\hat{\theta}_{ML}(F_0), F_0\right) = \arg \min_{F_0} \left\| \mathbf{x} - \mathbf{H}(F_0) \hat{\theta}_{ML}(F_0) \right\|^2 \\ &= \arg \min_{F_0} \left\| \mathbf{x} - \mathbf{H}(F_0) \mathbf{H}^\#(F_0) \mathbf{x} \right\|^2 \\ &= \arg \min_{F_0} \left\| \left( \mathbf{I} - \mathbf{H}(F_0) \mathbf{H}^\#(F_0) \right) \mathbf{x} \right\|^2 \\ &= \arg \min_{F_0} \left\| \left( \mathbf{I} - \mathbf{P}_H(F_0) \right) \mathbf{x} \right\|^2 \\ &= \arg \min_{F_0} \left\| \mathbf{P}_H^\perp(F_0) \mathbf{x} \right\|^2 \\ &= \arg \max_{F_0} \left\| \mathbf{P}_H(F_0) \mathbf{x} \right\|^2\end{aligned}$$

$$\hat{F}_{0,ML} = \hat{F}_{0,NLLS} = \arg \max_{F_0} \left\| \mathbf{P}_H(F_0) \mathbf{x} \right\|^2$$

- The ML/NLLS estimate of  $F_0$  is the value that maximizes the energy of the component of the data vector  $\mathbf{x}$  onto the subspace generated by the columns of matrix  $\mathbf{H}$ .
- It can be shown that this is equivalent to find the location of the maximum in the **Periodogram** of vector  $\mathbf{x}$ .
- The projection matrix of  $\mathbf{H}$  is given by:

$$\mathbf{P}_H = \mathbf{H}\mathbf{H}^\# = \begin{bmatrix} \mathbf{h}_1 & \mathbf{h}_2 \end{bmatrix} \cdot \frac{2}{N} \begin{bmatrix} \mathbf{h}_1^T \\ \mathbf{h}_2^T \end{bmatrix} = \frac{2}{N} \begin{bmatrix} \mathbf{h}_1 & \mathbf{h}_2 \end{bmatrix} \begin{bmatrix} \mathbf{h}_1^T \\ \mathbf{h}_2^T \end{bmatrix} = \mathbf{P}_H^T$$

$$\mathbf{P}_H(F_0) = \frac{2}{N} [\mathbf{h}_1 \quad \mathbf{h}_2] \begin{bmatrix} \mathbf{h}_1^T \\ \mathbf{h}_2^T \end{bmatrix}$$

$$\hat{F}_{0,ML} = \arg \max_{F_0} \left\| \mathbf{P}_H(F_0) \mathbf{x} \right\|^2$$

$$\begin{aligned} \left\| \mathbf{P}_H(F_0) \mathbf{x} \right\|^2 &= (\mathbf{P}_H \mathbf{x})^T (\mathbf{P}_H \mathbf{x}) = \mathbf{x}^T \mathbf{P}_H^T \mathbf{P}_H \mathbf{x} = \mathbf{x}^T \mathbf{P}_H \mathbf{P}_H \mathbf{x} = \mathbf{x}^T \mathbf{P}_H \mathbf{x} \\ &= \frac{2}{N} \mathbf{x}^T [\mathbf{h}_1 \quad \mathbf{h}_2] \begin{bmatrix} \mathbf{h}_1^T \\ \mathbf{h}_2^T \end{bmatrix} \mathbf{x} = \frac{2}{N} [\mathbf{x}^T \mathbf{h}_1 \quad \mathbf{x}^T \mathbf{h}_2] \begin{bmatrix} \mathbf{h}_1^T \mathbf{x} \\ \mathbf{h}_2^T \mathbf{x} \end{bmatrix} \\ &= \frac{2}{N} \begin{bmatrix} \mathbf{h}_1^T \mathbf{x} \\ \mathbf{h}_2^T \mathbf{x} \end{bmatrix}^T \begin{bmatrix} \mathbf{h}_1^T \mathbf{x} \\ \mathbf{h}_2^T \mathbf{x} \end{bmatrix} = \frac{2}{N} \left\| \begin{bmatrix} \mathbf{h}_1^T \mathbf{x} \\ \mathbf{h}_2^T \mathbf{x} \end{bmatrix} \right\|^2 = \frac{2}{N} \left( (\mathbf{h}_1^T \mathbf{x})^2 + (\mathbf{h}_2^T \mathbf{x})^2 \right) \\ &= \frac{2}{N} \left| (\mathbf{h}_1 - j\mathbf{h}_2)^T \mathbf{x} \right|^2 = \frac{2}{N} \left| \sum_{n=0}^{N-1} X_n (\cos(2\pi F_0 n) - j \sin(2\pi F_0 n)) \right|^2 \\ &= 2 \cdot \frac{1}{N} \left| \sum_{n=0}^{N-1} X_n e^{-j2\pi F_0 n} \right|^2 \rightarrow \text{Periodogram of } \mathbf{x} \end{aligned}$$

■ Once we have derived the NLLS estimate of  $F_0$  (which in AWGN coincides with the ML estimator), we can calculate  $\mathbf{H}$  and derive the ML/LS estimate of  $\theta$ :

$$\mathbf{H}(\hat{F}_{0,ML}) = \begin{bmatrix} \mathbf{h}_1(\hat{F}_{0,ML}) & \mathbf{h}_2(\hat{F}_{0,ML}) \end{bmatrix} = \begin{bmatrix} 1 & \cos(2\pi\hat{F}_{0,ML}) & \cdots & \cos(2\pi\hat{F}_{0,ML}(N-1)) \\ 0 & \sin(2\pi\hat{F}_{0,ML}) & \cdots & \sin(2\pi\hat{F}_{0,ML}(N-1)) \end{bmatrix}^T$$

$N \times 2$

$$\mathbf{H}^\#(\hat{F}_{0,ML}) = \left( \mathbf{H}^T(\hat{F}_{0,ML}) \mathbf{H}(\hat{F}_{0,ML}) \right)^{-1} \mathbf{H}^T(\hat{F}_{0,ML})$$

■ and finally, the ML estimates of  $A$  and  $\theta$ :

$$\hat{\boldsymbol{\theta}}_{ML} = \hat{\boldsymbol{\theta}}_{LS} = \mathbf{H}^\#(\hat{F}_{0,ML}) \mathbf{x} \Rightarrow \begin{cases} \hat{A}_{ML} = \sqrt{2B} \sqrt{\hat{\theta}_{1,ML}^2 + \hat{\theta}_{2,ML}^2} \\ \hat{\theta}_{ML} = -\arctg\left(\frac{\hat{\theta}_{2,ML}}{\hat{\theta}_{1,ML}}\right) \end{cases}$$



- It is **important** to stress that the location of the maximum of the **Periodogram** is no more the ML estimate of  $F_0$  if the noise is not Gaussian and/or not white, but it is still the NLLS estimate of  $F_0$ , whatever is the noise distribution and/or the noise color (i.e. the noise power spectral density).
- For the same reason, the output of the **Matched Filter** (matched to the know signal shape  $\mathbf{s}$ ) is no more the ML estimate of the amplitude  $A$  if the noise is not Gaussian and/or not white, but it is still the linear LS estimate of  $A$ , whatever is the noise distribution and/or the noise color (i.e. the noise power spectral density).



# **The Deterministic Linear Model in Colored Gaussian Noise**

■ **Assumption:** the noise is real, Gaussian, but correlated with known covariance matrix  $\mathbf{R}$ :

$$f_{\mathbf{x}}(\mathbf{x}; \boldsymbol{\theta}) = \frac{1}{\sqrt{(2\pi)^N |\mathbf{R}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T \mathbf{R}^{-1}(\mathbf{x} - \mathbf{H}\boldsymbol{\theta})\right)$$

■ The **log-Likelihood Function** is given by:

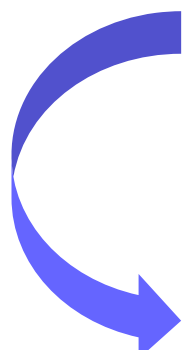
$$\ln L(\boldsymbol{\theta}) = \ln f_{\mathbf{x}}(\mathbf{x}; \boldsymbol{\theta}) = \kappa - \frac{1}{2}(\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T \mathbf{R}^{-1}(\mathbf{x} - \mathbf{H}\boldsymbol{\theta})$$

■ The **ML estimate** is obtained by solving the following problem:

$$\begin{aligned} \hat{\boldsymbol{\theta}}_{ML} &= \arg \max_{\boldsymbol{\theta}} \ln L(\boldsymbol{\theta}) = \arg \min_{\boldsymbol{\theta}} (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T \mathbf{R}^{-1}(\mathbf{x} - \mathbf{H}\boldsymbol{\theta}) \\ &= \arg \min_{\boldsymbol{\theta}} \|\mathbf{x} - \mathbf{H}\boldsymbol{\theta}\|_{\mathbf{R}^{-1}}^2 = \hat{\boldsymbol{\theta}}_{WLS} \quad \text{Weighted Least Squares (WLS)} \end{aligned}$$



- In general, the **WLS estimate** is given by:


$$\begin{aligned}\hat{\boldsymbol{\theta}}_{WLS} &= \arg \min_{\boldsymbol{\theta}} (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T \mathbf{W} (\mathbf{x} - \mathbf{H}\boldsymbol{\theta}) = \arg \min_{\boldsymbol{\theta}} \|\mathbf{x} - \mathbf{H}\boldsymbol{\theta}\|_{\mathbf{W}}^2 \\ &= \arg \min_{\boldsymbol{\theta}} \sum_{k=0}^{N-1} \sum_{r=0}^{N-1} w_{k,r} \left( x_k - \sum_{i=1}^p \theta_i h_{i,k} \right) \left( x_r - \sum_{i=1}^p \theta_i h_{i,r} \right)\end{aligned}$$

where  $x_k = [\mathbf{x}]_k$ ,  $\theta_i = [\boldsymbol{\theta}]_i$  and  $h_{i,k} = [\mathbf{H}]_{i,k}$

$$\Rightarrow \hat{\boldsymbol{\theta}}_{ML} = \hat{\boldsymbol{\theta}}_{WLS} \Big|_{\mathbf{W}=\mathbf{R}^{-1}}$$

- The weighting matrix **W** has the purpose to introduce different weights for different data, e.g. because some of them could be more noisy or less reliable.

■ *Example:*

$$\mathbf{R} = \begin{bmatrix} \sigma_0^2 & 0 & \cdots & 0 \\ 0 & \sigma_1^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{N-1}^2 \end{bmatrix} \Rightarrow \mathbf{W} = \mathbf{R}^{-1} = \begin{bmatrix} 1/\sigma_0^2 & 0 & \cdots & 0 \\ 0 & 1/\sigma_1^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1/\sigma_{N-1}^2 \end{bmatrix}$$


$$\hat{\boldsymbol{\theta}}_{WLS} = \arg \min_{\boldsymbol{\theta}} (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T \mathbf{W} (\mathbf{x} - \mathbf{H}\boldsymbol{\theta}) = \arg \min_{\boldsymbol{\theta}} \sum_{k=0}^{N-1} \frac{1}{\sigma_k^2} \left( \mathbf{x}_k - \sum_{i=1}^p \theta_i \mathbf{h}_{i,k} \right)^2$$

■ The more noisy data are weighted less, since they are less reliable.

- The **ML** estimate of  $\theta$  in correlated additive Gaussian noise is still **linear, unbiased, Gaussian, efficient** and **consistent**:

$$\hat{\theta}_{ML} = (\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{R}^{-1} \mathbf{X}$$

$$E\{\hat{\theta}_{ML}\} = (\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{R}^{-1} E\{\mathbf{X}\} = (\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{R}^{-1} \mathbf{H} \theta = \theta$$

$$MSE\{\hat{\theta}_{ML}\} = CRB(\theta) = (\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1}$$

$$\hat{\theta}_{ML} \in \mathcal{N}(\theta, CRB(\theta))$$

■ **Proof:** We can easily derive the **ML estimator** of  $\theta$  in **correlated noise** by applying the previous results to the “whitened” data  $\mathbf{Z}$ :

$$\text{Whitening : } \mathbf{Z} \triangleq \mathbf{R}^{-1/2} \mathbf{X} = \mathbf{R}^{-1/2} \mathbf{H} \theta + \mathbf{R}^{-1/2} \mathbf{W} = \tilde{\mathbf{H}} \theta + \tilde{\mathbf{W}} \Rightarrow \mathbf{Z} \in \mathcal{N}(\tilde{\mathbf{H}} \theta, \mathbf{I})$$

$$\begin{aligned} E\{\tilde{\mathbf{W}} \tilde{\mathbf{W}}^T\} &= E\{(\mathbf{R}^{-1/2} \mathbf{W})(\mathbf{R}^{-1/2} \mathbf{W})^T\} \\ &= E\left\{\mathbf{R}^{-1/2} \mathbf{W} \mathbf{W}^T (\mathbf{R}^{-1/2})^T\right\} \\ &= \mathbf{R}^{-1/2} E\{\mathbf{W} \mathbf{W}^T\} (\mathbf{R}^{-1/2})^T \\ &= \mathbf{R}^{-1/2} \mathbf{R} \mathbf{R}^{-1/2} \\ &= \mathbf{R}^{-1/2} \mathbf{R}^{1/2} \mathbf{R}^{1/2} \mathbf{R}^{-1/2} = \mathbf{I} \quad \rightarrow \quad \sigma_{\tilde{\mathbf{W}}}^2 = 1 \end{aligned}$$

$$\text{since } \mathbf{R} = \mathbf{R}^T \rightarrow \mathbf{R}^{-1/2} = (\mathbf{R}^{-1/2})^T$$

- The **ML estimator** of  $\theta$  for **correlated noise** is immediately derived:

$$\begin{aligned}
 \hat{\theta}_{ML} &= \tilde{\mathbf{H}}^{\#} \mathbf{Z} = \left( \tilde{\mathbf{H}}^T \tilde{\mathbf{H}} \right)^{-1} \tilde{\mathbf{H}}^T \mathbf{Z} \Big|_{\tilde{\mathbf{H}}=\mathbf{R}^{-1/2}\mathbf{H}, \mathbf{Z}=\mathbf{R}^{-1/2}\mathbf{X}} \\
 &= \left( \left( \mathbf{R}^{-1/2} \mathbf{H} \right)^T \mathbf{R}^{-1/2} \mathbf{H} \right)^{-1} \left( \mathbf{R}^{-1/2} \mathbf{H} \right)^T \mathbf{R}^{-1/2} \mathbf{X} \\
 &= \left( \mathbf{H}^T \left( \mathbf{R}^{-1/2} \right)^T \mathbf{R}^{-1/2} \mathbf{H} \right)^{-1} \mathbf{H}^T \left( \mathbf{R}^{-1/2} \right)^T \mathbf{R}^{-1/2} \mathbf{X} \\
 &= \left( \mathbf{H}^T \mathbf{R}^{-1/2} \mathbf{R}^{-1/2} \mathbf{H} \right)^{-1} \mathbf{H}^T \mathbf{R}^{-1/2} \mathbf{R}^{-1/2} \mathbf{X} \\
 &= \left( \mathbf{H}^T \mathbf{R}^{-1} \mathbf{H} \right)^{-1} \mathbf{H}^T \mathbf{R}^{-1} \mathbf{X}
 \end{aligned}$$

- In a similar way, we can derive that the MSE is given by:

$$MSE\{\hat{\theta}_{ML}\} = \sigma_{\tilde{W}}^2 \left( \tilde{\mathbf{H}}^T \tilde{\mathbf{H}} \right)^{-1} \Big|_{\tilde{\mathbf{H}}=\mathbf{R}^{-1/2}\mathbf{H}, \sigma_{\tilde{W}}^2=1} = \left( \mathbf{H}^T \mathbf{R}^{-1/2} \mathbf{R}^{-1/2} \mathbf{H} \right)^{-1} = \left( \mathbf{H}^T \mathbf{R}^{-1} \mathbf{H} \right)^{-1}$$

■ The estimator we previously derived is also called the **Best Linear Unbiased Estimator (BLUE)**, since it can be derived for the same estimation problem (without the assumption of Gaussianity for the noise  $\mathbf{W}$ ), minimizing the MSE under the constraint that the estimator is linear and unbiased:

$$\mathbf{X} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W}, \quad \hat{\boldsymbol{\theta}}_{BLUE} = (\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{R}^{-1} \mathbf{X}$$

$$E\{\hat{\boldsymbol{\theta}}_{BLUE}\} = \boldsymbol{\theta}, \quad MSE\{\hat{\boldsymbol{\theta}}_{BLUE}\} = (\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1}$$

■ The **Gauss-Markov theorem** states that when  $\mathbf{X}$  obeys the linear-Gaussian model, i.e.  $\mathbf{X} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W}$  and  $\mathbf{W} \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$ , then the above BLUE estimator is also the **Minimum Variance Unbiased Estimator (MVUE)**.

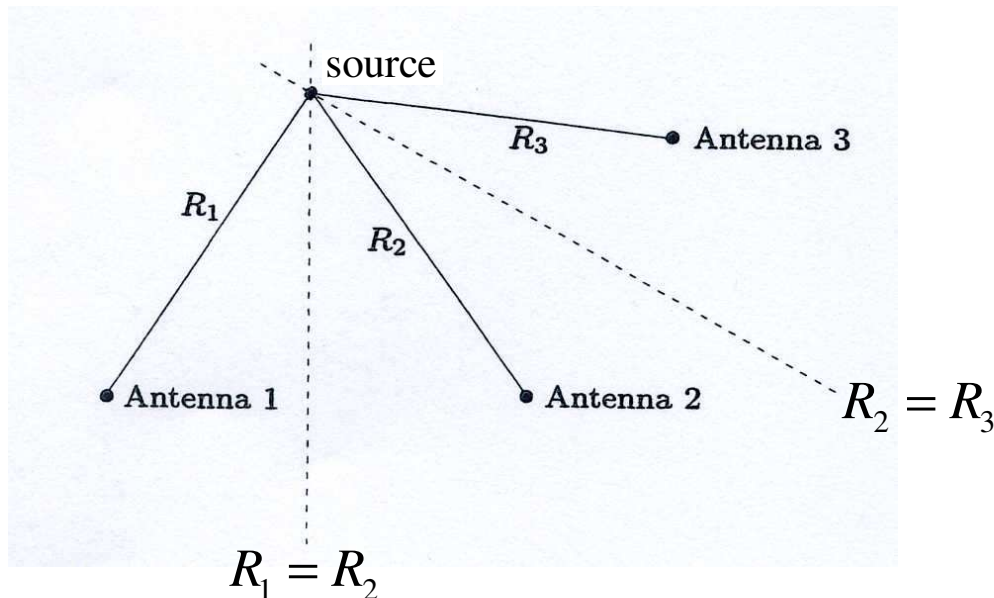
■ Note that when the noise is non Gaussian, it may exist a nonlinear estimator that has lower MSE than the BLUE.

## Example: Source Localization

79

■ **Problem:** determine the position of a source based on the emitted signal of the source.

■ A common estimation method is based on **Time Difference of Arrival (TDOA)** measurements.



*Example :*

$$\left. \begin{array}{l} t_2 - t_1 = 0 \\ t_3 - t_2 = 0 \end{array} \right\} \Rightarrow \left\{ \begin{array}{l} R_1 = R_2 \\ R_2 = R_3 \end{array} \right.$$

The intersection of these 2 lines gives the source position.

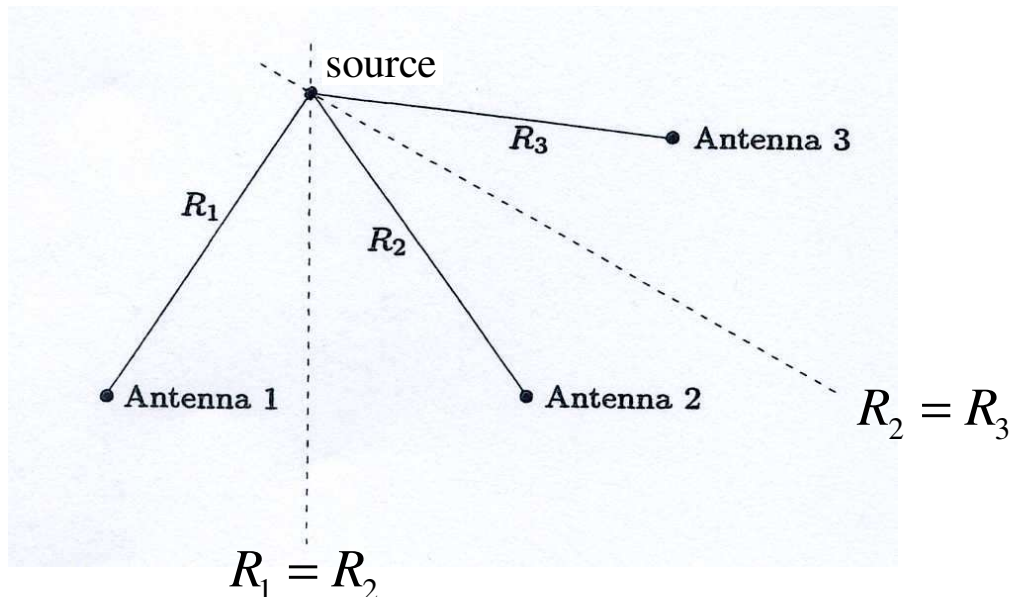
$t_i$  = Time of Arrival (of the signal emitted by the source) at receive antenna  $i$

■ The ToA  $t_i$  can be measured (i.e. estimated) as the location of the peak of the Matched Filter output (as described in Ch.3, p.230) or the Energy Detector (ED).

## Example: Source Localization

80

- **Problem:** determine the position of a source based on the emitted signal of the source.
- A common estimation method is based on **Time Difference of Arrival (TDOA)** measurements.



In the more general situation for which the TDOAs are not zero, the dashed lines become **hyperbolas**, the intersection again giving the source position.

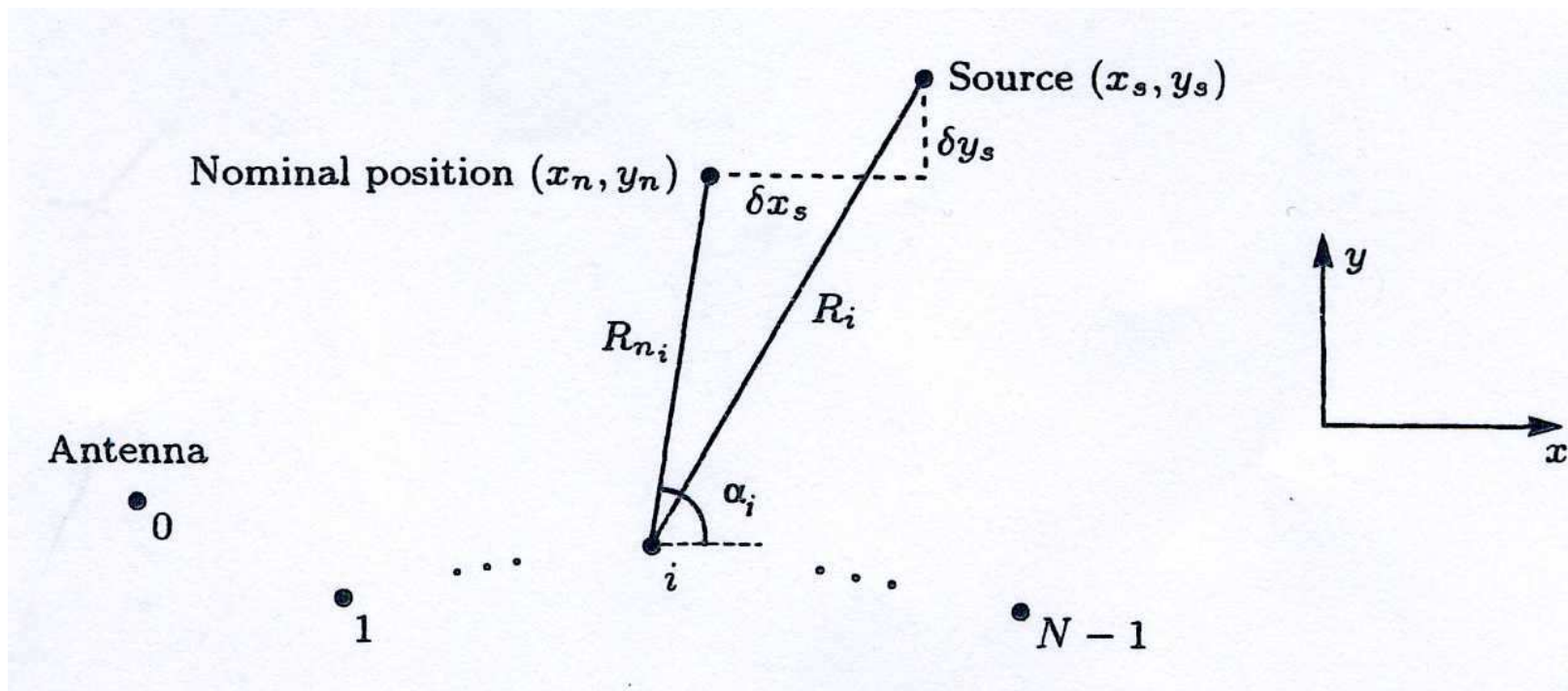
- We need at least 3 receive sensors for localization, and when the noise is present in the measurements, it is desirable to have more.



## Example: Source Localization

81

- Assume that  $N$  receive sensors (e.g. receive antennas) have been placed at known locations and that the **Time of Arrival (TOA)** measurements  $t_i$  for  $i=0,1,2, \dots, N-1$ , are available and assume that the source and the receive sensors are on the same plane (2D problem).
- The problem is to estimate the source position  $(x_s, y_s)$  from the observation of the  $N$  arrival times  $\{t_i; i=0,1, \dots, N-1\}$ .



- The (measured) times of arrival  $\{t_i\}$  are assumed to be corrupted by noise with zero mean and known covariance but otherwise unknown distribution.
- For a signal emitted by the source at time  $t=T_0$ , the ToA measurements are modeled by:

$$t_i = T_0 + \frac{R_i}{c} + e_i, \quad i = 0, 1, 2, \dots, N-1$$

- where  $e_i$  are the measurement noises and  $c$  denotes the propagation speed.
- The noise samples are assumed to be zero mean with variance  $\sigma^2$  and uncorrelated with each other. No assumptions are made about the pdf of the noise.
- Knowledge of  $T_0$  would require accurate clock synchronization between the source and the receivers. Often for economic reasons this is avoided. We assume here that the time  $T_0$  when the source emits a signal is unknown.

## Example: Source Localization

83

- Unknown position vector:  $\mathbf{\theta} = [x_s \quad y_s]^T$
- Position of the  $i$ th receive sensor:  $\mathbf{p}_i = [x_i \quad y_i]^T$
- Range between source and receive sensor  $i$ th:

$$R_i = \|\mathbf{\theta} - \mathbf{p}_i\| = \sqrt{(x_s - x_i)^2 + (y_s - y_i)^2}, \quad i = 0, 1, 2, \dots, N-1$$

- To remove the dependency of the observed data on the unknown time  $T_0$ , it is customary to consider difference of arrivals or TDOA measurements:

$$\tau_i = t_i - t_{i-1} = \frac{R_i - R_{i-1}}{c} + e_i - e_{i-1} = \frac{R_i - R_{i-1}}{c} + w_i, \quad i = 1, 2, \dots, N-1$$

$$= \frac{\sqrt{(x_s - x_i)^2 + (y_s - y_i)^2} - \sqrt{(x_s - x_{i-1})^2 + (y_s - y_{i-1})^2}}{c} + w_i$$

### ■ TDOA data model:

$$\underset{(N-1) \times 1}{\boldsymbol{\tau}} = \mathbf{s}(\boldsymbol{\theta}) + \mathbf{w}, \quad \boldsymbol{\theta} = \begin{bmatrix} x_s \\ y_s \end{bmatrix}$$

$$s_i(\boldsymbol{\theta}) = \frac{R_i - R_{i-1}}{c} = \frac{\sqrt{(x_s - x_i)^2 + (y_s - y_i)^2} - \sqrt{(x_s - x_{i-1})^2 + (y_s - y_{i-1})^2}}{c}$$

■ To estimate the unknown parameter vector  $\boldsymbol{\theta}$  we could use the **Non Linear Least Squares (NLLS)** method:

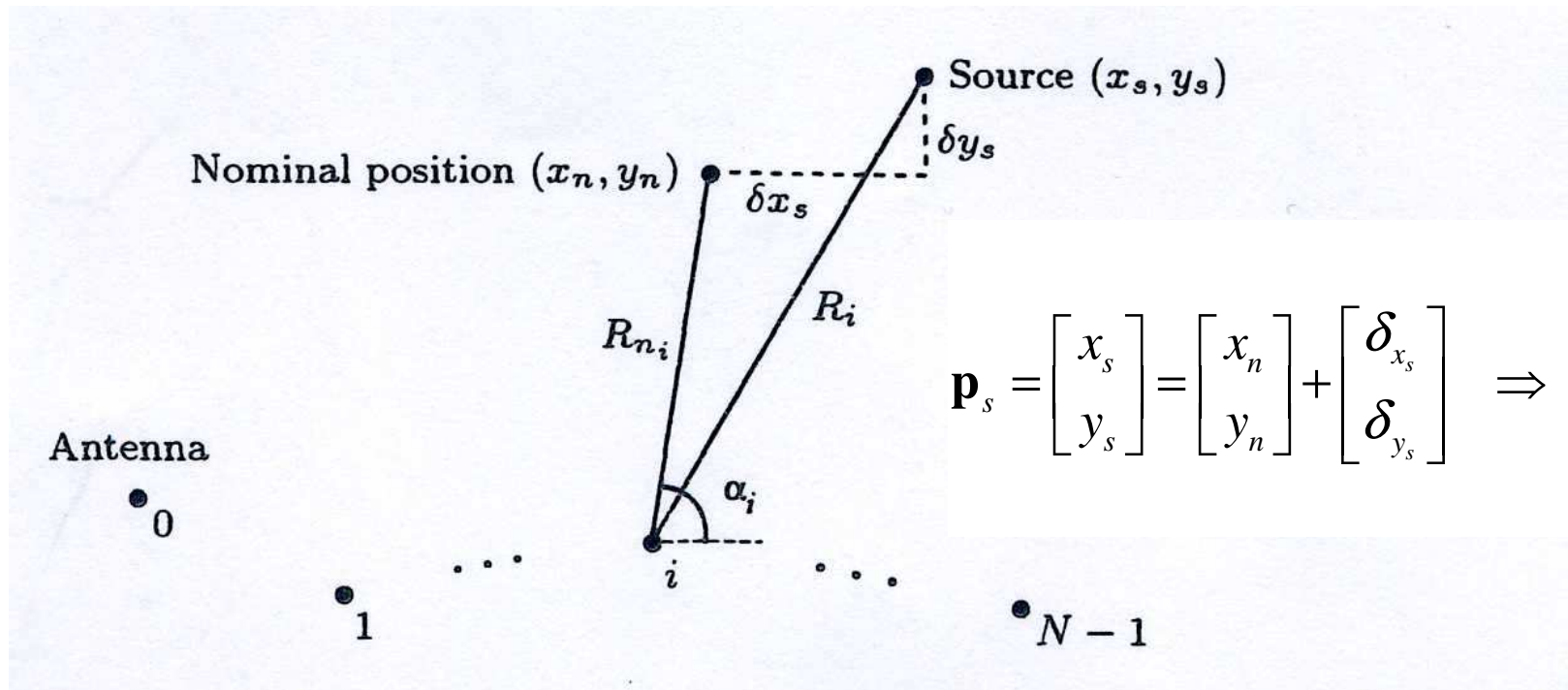
$$\hat{\boldsymbol{\theta}}_{NLLS} = \arg \min_{\boldsymbol{\theta}} \|\boldsymbol{\tau} - \mathbf{s}(\boldsymbol{\theta})\|^2$$

■ The NLLS estimator does not require statistical characterization of the noise vector  $\mathbf{w}$ , but it is computationally heavy (it requires iterative optimization techniques).

## Example: Source Localization

85

- Let us now assume that a **nominal source position**  $(x_n, y_n)$  is available that is close to the true source position.
- Such a situation is typical when a source is being tracked; in that case the nominal position is the position of the source at the previous measurement time.



$$R_{n_i} = \sqrt{(x_n - x_i)^2 + (y_n - y_i)^2}, \quad i = 0, 1, 2, \dots, N-1$$

- Range between source and receive sensor  $i$ th as a function of the nominal position:

$$R_i = \sqrt{(x_s - x_i)^2 + (y_s - y_i)^2} = \sqrt{(x_n + \delta_{x_s} - x_i)^2 + (y_n + \delta_{y_s} - y_i)^2}$$

- If we now use a first-order Taylor expansion of  $R_i$ , considered as a 2D function of the source position  $(x_s, y_s)$ , about the nominal position  $(x_n, y_n)$ , we get:

$$R_i \cong R_{n_i} + \frac{x_n - x_i}{R_{n_i}} \delta_{x_s} + \frac{y_n - y_i}{R_{n_i}} \delta_{y_s} = R_{n_i} + \cos(\alpha_i) \delta_{x_s} + \sin(\alpha_i) \delta_{y_s}$$



$$\tau_i = \frac{R_i - R_{i-1}}{c} + w_i \cong \frac{R_{n_i} - R_{n_{i-1}}}{c} + \frac{\cos(\alpha_i) - \cos(\alpha_{i-1})}{c} \delta_{x_s} + \frac{\sin(\alpha_i) - \sin(\alpha_{i-1})}{c} \delta_{y_s} + w_i$$

## ■ Linearized measurement data model:

$$z_i = \tau_i - \frac{R_{n_i} - R_{n_{i-1}}}{c} = \frac{\cos \alpha_i - \cos \alpha_{i-1}}{c} \delta_{x_s} + \frac{\sin \alpha_i - \sin \alpha_{i-1}}{c} \delta_{y_s} + w_i$$

Known! It can be incorporated in the measurement vector.

$$\underset{(N-1) \times 1}{\mathbf{z}} = \mathbf{H} \underset{2 \times 1}{\boldsymbol{\theta}} + \underset{1 \times 1}{\mathbf{w}}, \quad \boldsymbol{\theta} = \begin{bmatrix} \delta_{x_s} \\ \delta_{y_s} \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_{N-1} \end{bmatrix} = \begin{bmatrix} e_1 - e_0 \\ \vdots \\ e_{N-1} - e_{N-2} \end{bmatrix}$$

$$\underset{(N-1) \times 2}{\mathbf{H}} = \frac{1}{c} \begin{bmatrix} \cos \alpha_1 - \cos \alpha_0 & \sin \alpha_1 - \sin \alpha_0 \\ \cos \alpha_2 - \cos \alpha_1 & \sin \alpha_2 - \sin \alpha_1 \\ \vdots & \vdots \\ \cos \alpha_{N-1} - \cos \alpha_{N-2} & \sin \alpha_{N-1} - \sin \alpha_{N-2} \end{bmatrix}$$

## Example: Source Localization

88

- The noise vector  $\mathbf{w}$  is zero mean, correlated, with unknown distribution:

$$\underset{(N-1) \times 1}{\mathbf{w}} = \begin{bmatrix} w_1 \\ \vdots \\ w_{N-1} \end{bmatrix} = \begin{bmatrix} e_1 - e_0 \\ \vdots \\ e_{N-1} - e_{N-2} \end{bmatrix} = \underbrace{\begin{bmatrix} -1 & 1 & 0 & 0 & \cdots & 0 \\ 0 & -1 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & -1 & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} e_0 \\ e_1 \\ \vdots \\ e_{N-1} \end{bmatrix}}_{\mathbf{e}} = \underset{(N-1) \times N \quad N \times 1}{\mathbf{A}} \underset{N \times 1}{\mathbf{e}}$$

- The covariance matrix of the noise vector  $\mathbf{w}$  can be expressed as a function of the covariance matrix of the noise vector  $\mathbf{e}$  in the TOA measurements (that we assumed to be uncorrelated):

$$\begin{aligned} \mathbf{C}_w &= E\{\mathbf{w}\mathbf{w}^T\} = E\{(\mathbf{A}\mathbf{e})(\mathbf{A}\mathbf{e})^T\} = E\{\mathbf{A}\mathbf{e}\mathbf{e}^T\mathbf{A}^T\} = \mathbf{A}E\{\mathbf{e}\mathbf{e}^T\}\mathbf{A}^T \\ &= \mathbf{A}\mathbf{C}_e\mathbf{A}^T = \mathbf{A}(\sigma^2\mathbf{I})\mathbf{A}^T = \sigma^2\mathbf{A}\mathbf{A}^T \end{aligned}$$



■ Thanks to the linearization, the problem has been recast as the problem of estimating the vector  $\boldsymbol{\theta}$  based on the observation of the  $(N-1)$ -dimensional vector  $\mathbf{z}$ , when  $\mathbf{H}$  is a-priori known and  $\mathbf{w}$  is zero mean, correlated, with known covariance matrix  $\mathbf{C}_w$ :

$$\underset{(N-1) \times 1}{\mathbf{z}} = \mathbf{H} \boldsymbol{\theta} + \mathbf{w}, \quad \boldsymbol{\theta} = \begin{bmatrix} \delta_{x_s} \\ \delta_{y_s} \end{bmatrix}$$

$$\underset{(N-1) \times 2}{\mathbf{H}} = \frac{1}{c} \begin{bmatrix} \cos \alpha_1 - \cos \alpha_0 & \sin \alpha_1 - \sin \alpha_0 \\ \cos \alpha_2 - \cos \alpha_1 & \sin \alpha_2 - \sin \alpha_1 \\ \vdots & \vdots \\ \cos \alpha_{N-1} - \cos \alpha_{N-2} & \sin \alpha_{N-1} - \sin \alpha_{N-2} \end{bmatrix}$$

■ Since we do not know the PDF of  $\mathbf{w}$  we cannot derive the ML estimator. We can use the BLUE, which coincides with the WLS with weighting matrix equal to the inverse of the noise covariance matrix.

### ■ BLUE/WLS estimator and MSE matrix:

$$\hat{\boldsymbol{\theta}}_{BLUE} = \hat{\boldsymbol{\theta}}_{WLS} \Big|_{\mathbf{W}=\mathbf{C}_w^{-1}} = \left( \mathbf{H}^T \mathbf{C}_w^{-1} \mathbf{H} \right)^{-1} \mathbf{H}^T \mathbf{C}_w^{-1} \mathbf{z} = \left( \mathbf{H}^T \left( \mathbf{A} \mathbf{A}^T \right)^{-1} \mathbf{H} \right)^{-1} \mathbf{H}^T \left( \mathbf{A} \mathbf{A}^T \right)^{-1} \mathbf{z}$$

$$E \left\{ \hat{\boldsymbol{\theta}}_{BLUE} \right\} = \boldsymbol{\theta}$$

$$MSE \left\{ \hat{\boldsymbol{\theta}}_{BLUE} \right\} = \left( \mathbf{H}^T \mathbf{C}_w^{-1} \mathbf{H} \right)^{-1} = \sigma^2 \left( \mathbf{H}^T \left( \mathbf{A} \mathbf{A}^T \right)^{-1} \mathbf{H} \right)^{-1}$$

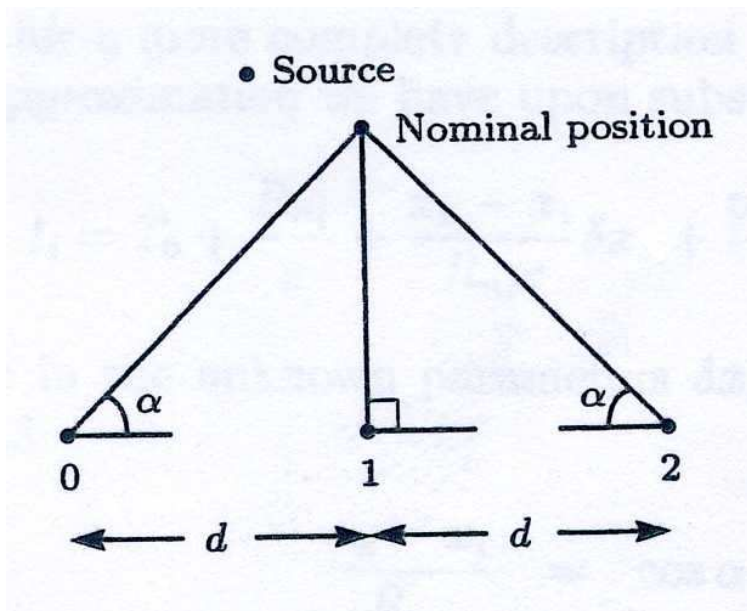
■ If the noise vector is Gaussian distributed, this estimator is also the ML estimator and it is efficient.

■ An estimator based on TDOA measurements was used in the terrestrial navigation system called **LORAN (Long Range Navigation)** but it is also widely adopted for indoor localization.

# Example: Source Localization

91

- **Example:**  $N=3$  elements Uniform Linear Array (ULA) line array (3 is the minimum number of antennas required):



$d$  is the so-called array **baseline**

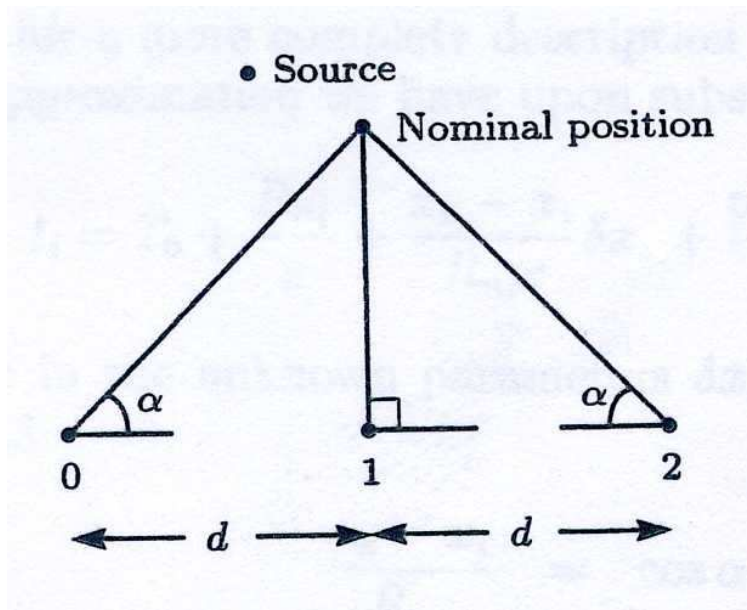
$$\mathbf{H}_{2 \times 2} = \frac{1}{c} \begin{bmatrix} -\cos \alpha & 1 - \sin \alpha \\ -\cos \alpha & -(1 - \sin \alpha) \end{bmatrix}$$

$$\mathbf{A}_{2 \times 3} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$MSE\{\hat{\boldsymbol{\theta}}_{BLUE}\} = \sigma^2 c^2 \begin{bmatrix} \frac{1}{2 \cos^2 \alpha} & 0 \\ 0 & \frac{3}{2(1 - \sin \alpha)^2} \end{bmatrix}$$

## Example: Source Localization

92



$d$  is the so-called array **baseline**

$$MSE\{\hat{\boldsymbol{\theta}}_{BLUE}\} = \sigma^2 c^2 \begin{bmatrix} \frac{1}{2\cos^2 \alpha} & 0 \\ 0 & \frac{3}{2(1-\sin \alpha)^2} \end{bmatrix}$$

■ For best localization we would like the angle  $\alpha$  to be small. This is achieved by making the **baseline**  $d$  large, so making large the **array aperture**, i.e. the total length. Note also that the localization accuracy is range dependent, with the best accuracy for short ranges or for  $\alpha$  small.

■ Let us now compare the performance of the **ML/BLUE** and **LS** estimators of  $\theta$  in additive correlated Gaussian noise.

■ Both are **linear** and **unbiased**:

$$\mathbf{X} = \mathbf{H}\theta + \mathbf{W} \rightarrow \mathbf{X} \in \mathcal{N}(\mathbf{H}\theta, \mathbf{R})$$

$$\left\{ \begin{array}{l} \hat{\theta}_{ML} = \hat{\theta}_{BLUE} = (\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{R}^{-1} \mathbf{X} \\ E\{\hat{\theta}_{ML}\} = E\{\hat{\theta}_{BLUE}\} = \theta \\ MSE\{\hat{\theta}_{ML}\} = MSE\{\hat{\theta}_{BLUE}\} = (\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1} \quad [ = CRB(\theta) \text{ if } \mathbf{W} \text{ is Gaussian } ] \end{array} \right.$$



$$\mathbf{X} = \mathbf{H}\boldsymbol{\theta} + \mathbf{W} \rightarrow \mathbf{X} \in \mathcal{N}(\mathbf{H}\boldsymbol{\theta}, \mathbf{R})$$

$$\left\{ \begin{array}{l} \hat{\boldsymbol{\theta}}_{LS} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{X} \\ \\ E\{\hat{\boldsymbol{\theta}}_{LS}\} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T E\{\mathbf{X}\} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{H}\boldsymbol{\theta} = \boldsymbol{\theta} \\ \\ MSE\{\hat{\boldsymbol{\theta}}_{LS}\} = E\{(\hat{\boldsymbol{\theta}}_{LS} - \boldsymbol{\theta})(\hat{\boldsymbol{\theta}}_{LS} - \boldsymbol{\theta})^T\} = E\{(\mathbf{H}^\# \mathbf{X} - \boldsymbol{\theta})(\mathbf{H}^\# \mathbf{X} - \boldsymbol{\theta})^T\} \\ \quad = E\{(\mathbf{H}^\# \mathbf{H}\boldsymbol{\theta} + \mathbf{H}^\# \mathbf{W} - \boldsymbol{\theta})(\mathbf{H}^\# \mathbf{H}\boldsymbol{\theta} + \mathbf{H}^\# \mathbf{W} - \boldsymbol{\theta})^T\} \\ \quad = E\{(\mathbf{H}^\# \mathbf{W})(\mathbf{H}^\# \mathbf{W})^T\} = \mathbf{H}^\# E\{\mathbf{W}\mathbf{W}^T\}(\mathbf{H}^\#)^T \\ \quad = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{R} [(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T]^T \\ \quad = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{R} \mathbf{H} (\mathbf{H}^T \mathbf{H})^{-1} \end{array} \right.$$

■ **Example:** Let us consider the previous problem, where we want to estimate amplitude and phase of a sinusoid:

$$\underset{N \times 2}{\mathbf{H}} = [\mathbf{h}_1 \quad \mathbf{h}_2] = \begin{bmatrix} 1 & \cos(2\pi F_0) & \cdots & \cos(2\pi F_0(N-1)) \\ 0 & \sin(2\pi F_0) & \cdots & \sin(2\pi F_0(N-1)) \end{bmatrix}^T$$

■ and assume that the Autocorrelation Function (ACF) of the samples has an exponential shape, i.e.  $W[n]$  is an AR(1) process, a.k.a. Markov process:

$$[\mathbf{R}]_{i,k} = \sigma_W^2 \rho^{|i-k|}, \quad i, k = 1, 2, \dots, N$$

■ where  $\rho$  is the one-lag correlation coefficient:  $-1 < \rho < 1$

■ If  $\rho > 0$ ,  $W[n]$  is a low-pass process, if  $\rho < 0$  it is an high-pass process.

■ If  $\rho = 0 \rightarrow \mathbf{R} = \sigma_W^2 \mathbf{I} \rightarrow \text{ML} = \text{LS}$

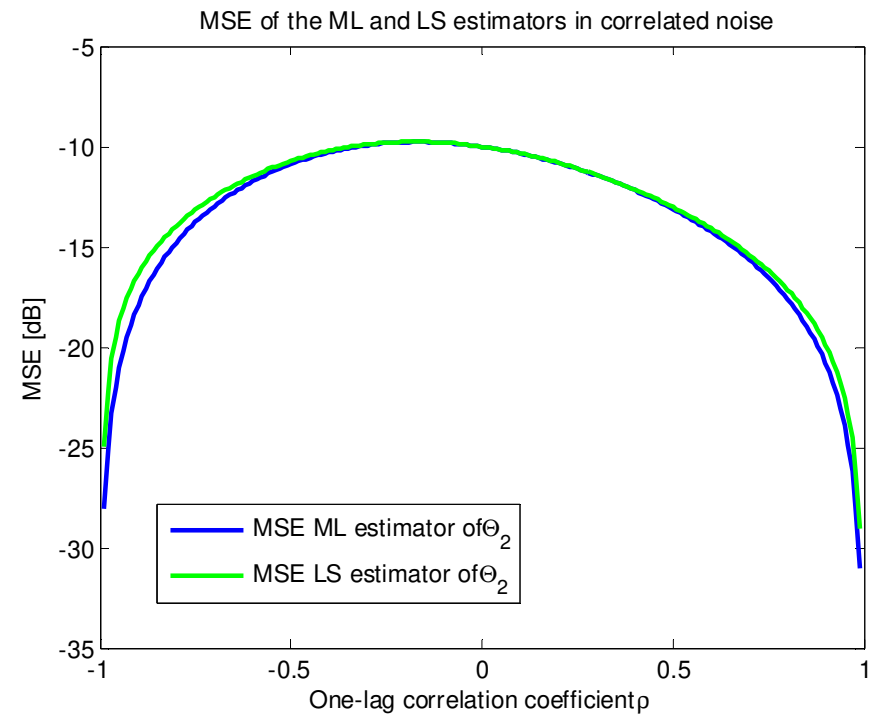
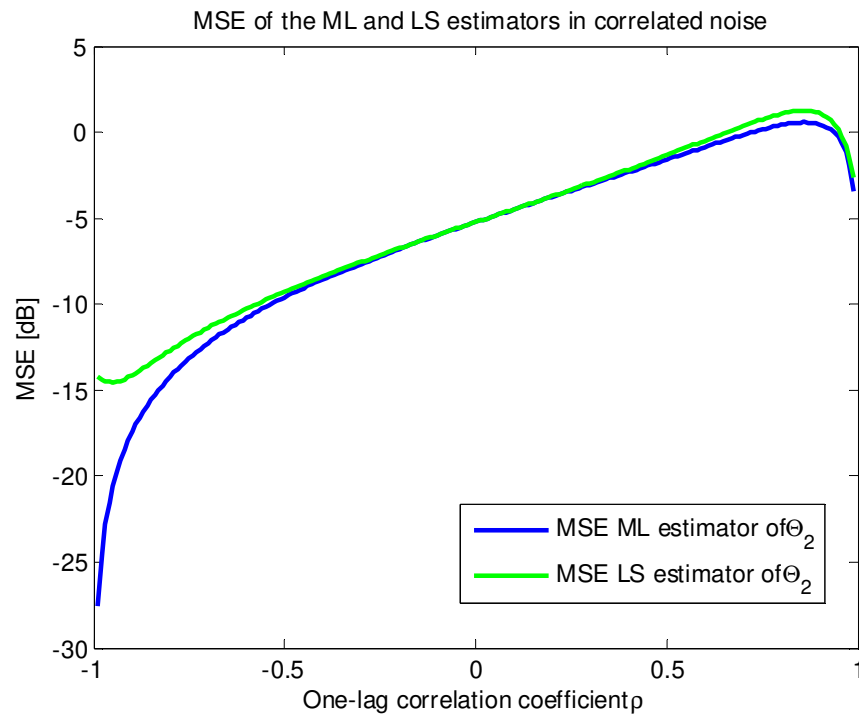


# The Deterministic Linear Model in CGN

96

$$F_0 = 0.01, \quad N = 20, \quad \sigma_w^2 = 1$$

$$F_0 = 0.3, \quad N = 20, \quad \sigma_w^2 = 1$$



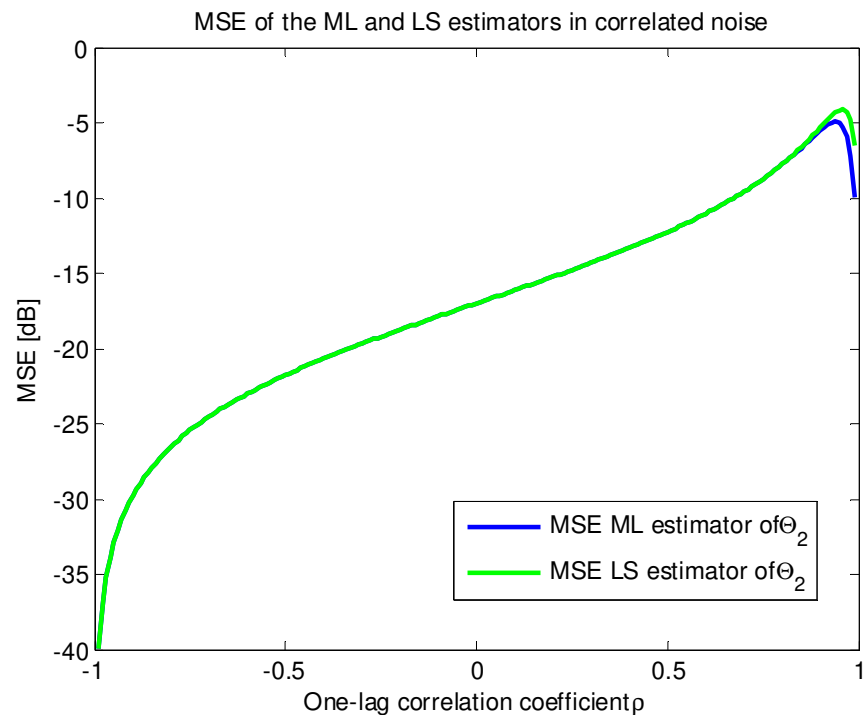
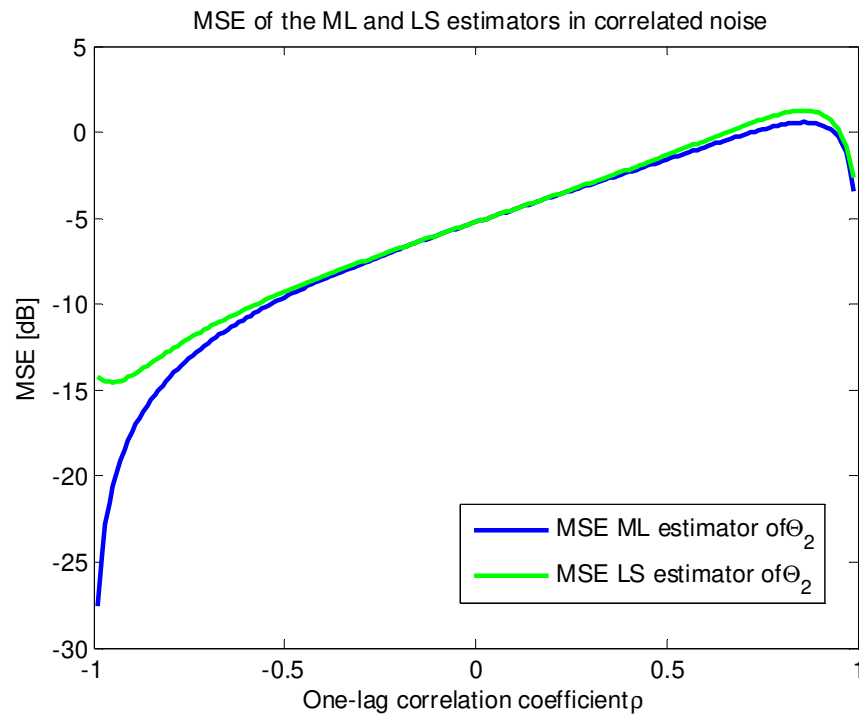


# The Deterministic Linear Model in CGN

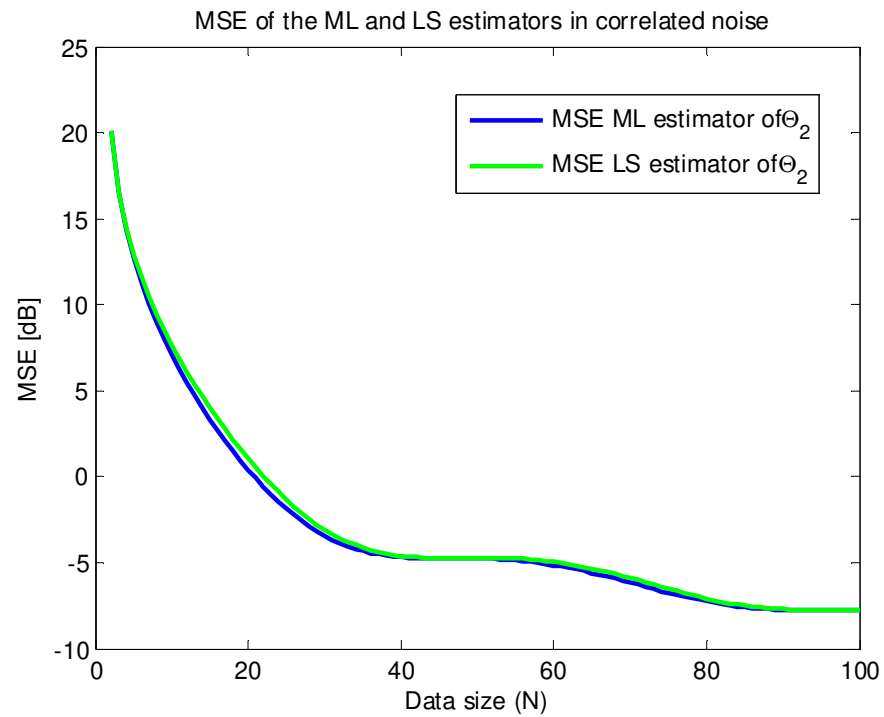
97

$$F_0 = 0.01, \quad N = 20, \quad \sigma_w^2 = 1$$

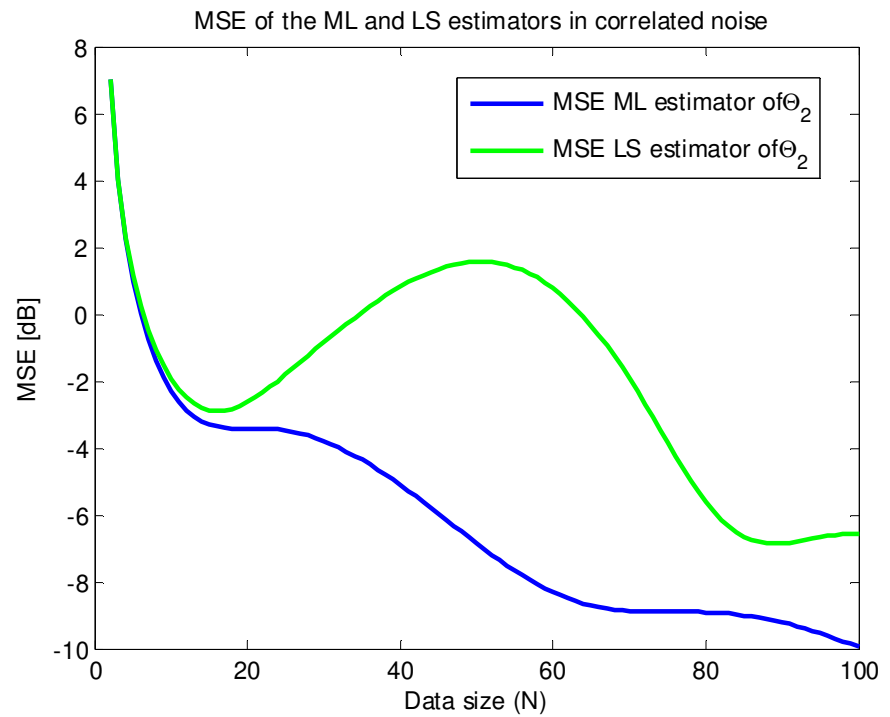
$$F_0 = 0.01, \quad N = 100, \quad \sigma_w^2 = 1$$



$$F_0 = 0.01, \quad \rho = 0.8, \quad \sigma_w^2 = 1$$



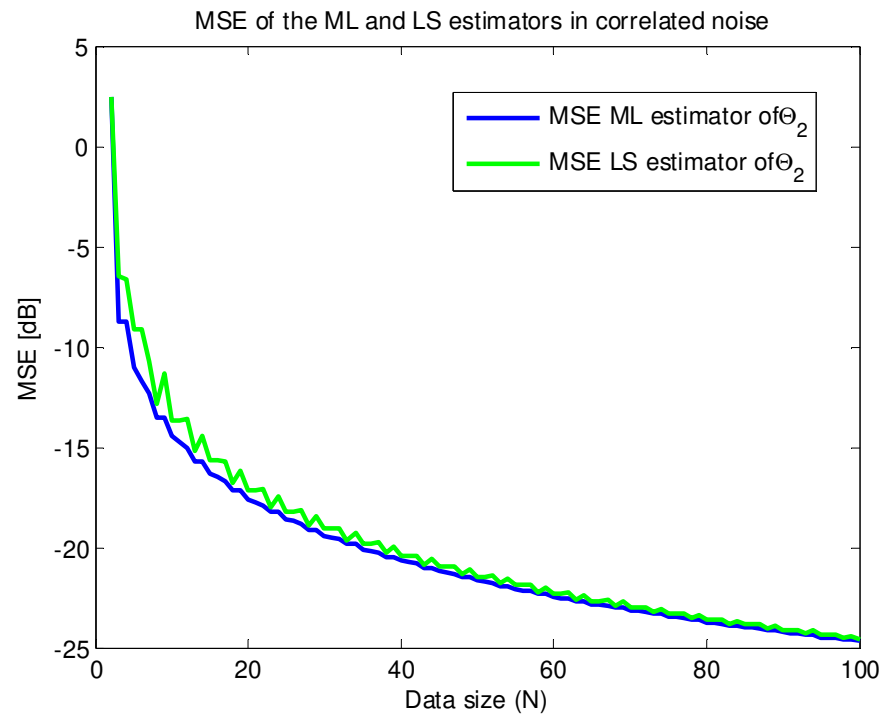
$$F_0 = 0.01, \quad \rho = 0.99, \quad \sigma_w^2 = 1$$



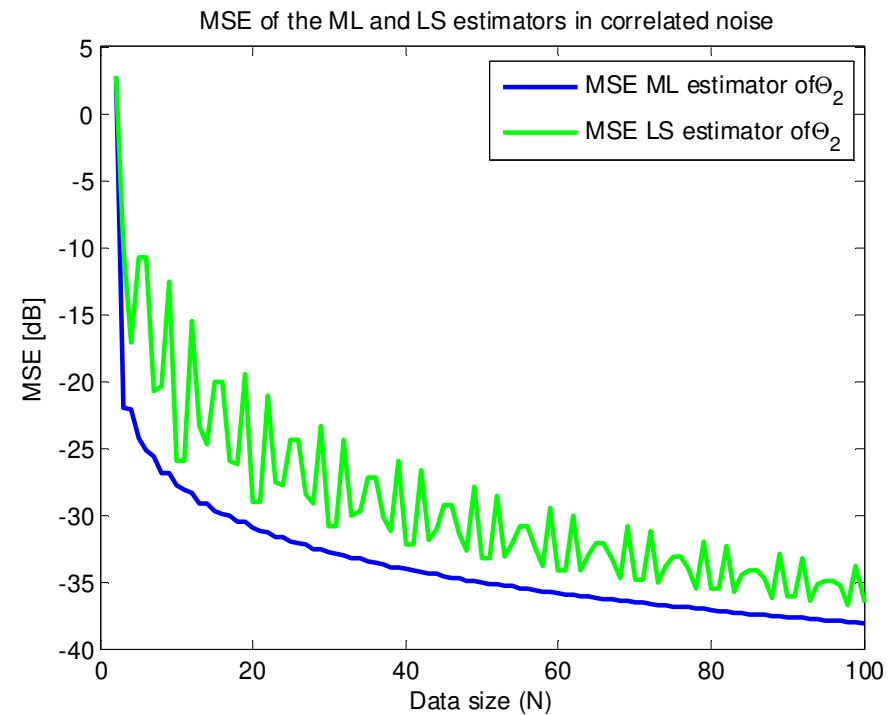
# The Deterministic Linear Model in CGN

99

$$F_0 = 0.3, \quad \rho = 0.8, \quad \sigma_w^2 = 1$$



$$F_0 = 0.3, \quad \rho = 0.99, \quad \sigma_w^2 = 1$$



$$\mathbf{X} = \mathbf{A}\mathbf{p} + \mathbf{W}$$

Estimate the signal amplitude:  $\theta = A$ , and  $\mathbf{H} = \mathbf{p}$ ,

$\mathbf{p}$  = signal vector, perfectly known,  $\mathbf{W} \in \mathcal{N}(0, \mathbf{R})$

$$\begin{aligned}\hat{A}_{ML} &= \hat{A}_{BLUE} = (\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{R}^{-1} \mathbf{X} = (\mathbf{p}^T \mathbf{R}^{-1} \mathbf{p})^{-1} \mathbf{p}^T \mathbf{R}^{-1} \mathbf{X} \\ &= \frac{\mathbf{p}^T \mathbf{R}^{-1} \mathbf{X}}{\mathbf{p}^T \mathbf{R}^{-1} \mathbf{p}} \quad \leftarrow \text{Whitening Matched Filter (WMF)}\end{aligned}$$

$$= \mathbf{h}^T \mathbf{X}, \quad \text{where } \mathbf{h} \triangleq \frac{\mathbf{R}^{-1} \mathbf{p}}{\mathbf{p}^T \mathbf{R}^{-1} \mathbf{p}} \text{ is the impulse response of the WMF}$$

■ This is the detection algorithm proposed by Brennan and Reed for coherent detectors when the signal shape is *a priori* known and the noise is additive, Gaussian, and correlated with known covariance matrix.

■ If the additive noise is correlated, with known covariance matrix  $\mathbf{R}$ , the ML estimate of the signal amplitude can be interpreted as the cascade of a **Whitening Filter (WF)** followed by a **Matched Filter (MF)**, matched to the “whitened” signal vector.

$$\mathbf{X} = A\mathbf{p} + \mathbf{W} \rightarrow \mathbf{X} \in \mathcal{N}(A\mathbf{p}, \mathbf{R})$$

$$\begin{cases} \boldsymbol{\eta} \triangleq E\{\mathbf{X}\} = E\{A\mathbf{p} + \mathbf{W}\} = A\mathbf{p} \\ \mathbf{R} \triangleq E\{(\mathbf{X} - \boldsymbol{\eta})(\mathbf{X} - \boldsymbol{\eta})^T\} = E\{\mathbf{W}\mathbf{W}^T\} = \begin{cases} \sigma_w^2 \mathbf{I} & (\text{AWGN}) \\ \mathbf{R} \neq \sigma_w^2 \mathbf{I} & (\text{AGN}) \end{cases} \end{cases}$$

$$AWGN: \hat{A}_{ML} = \hat{A}_{LS} = \frac{\mathbf{p}^T \mathbf{X}}{\mathbf{p}^T \mathbf{p}} = \frac{1}{E_p} \mathbf{p}^T \mathbf{X} \leftarrow \text{Matched Filter (MF)}$$

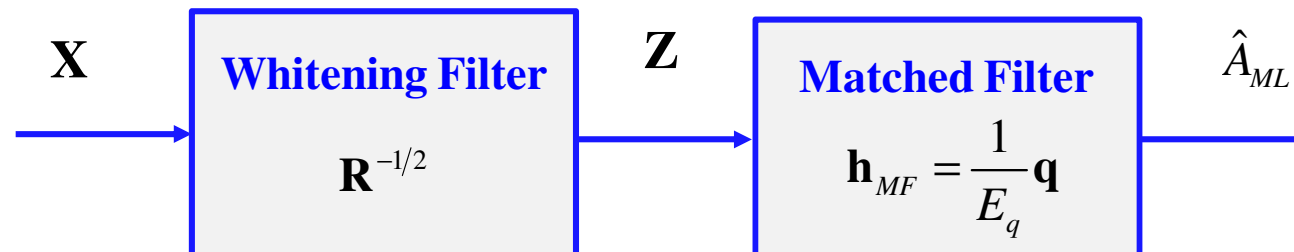
$$AGN: \hat{A}_{ML} = \hat{A}_{BLUE} = \frac{\mathbf{p}^T \mathbf{R}^{-1} \mathbf{X}}{\mathbf{p}^T \mathbf{R}^{-1} \mathbf{p}} \leftarrow \text{Whitening Matched Filter (WMF)}$$

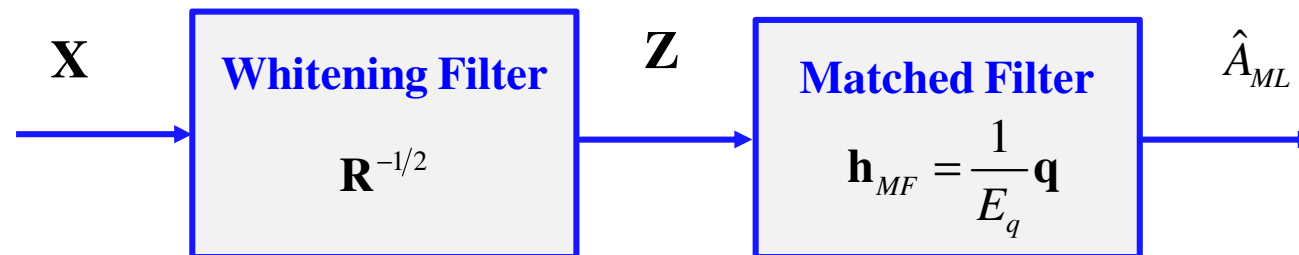
$$\hat{A}_{ML} = \frac{\mathbf{p}^T \mathbf{R}^{-1} \mathbf{X}}{\mathbf{p}^T \mathbf{R}^{-1} \mathbf{p}} = \frac{\mathbf{p}^T \mathbf{R}^{-1/2} \mathbf{R}^{-1/2} \mathbf{X}}{\mathbf{p}^T \mathbf{R}^{-1/2} \mathbf{R}^{-1/2} \mathbf{p}} = \frac{(\mathbf{R}^{-1/2} \mathbf{p})^T (\mathbf{R}^{-1/2} \mathbf{X})}{(\mathbf{R}^{-1/2} \mathbf{p})^T (\mathbf{R}^{-1/2} \mathbf{p})} = \frac{\mathbf{q}^T \mathbf{Z}}{\mathbf{q}^T \mathbf{q}} = \frac{1}{E_q} \mathbf{q}^T \mathbf{Z}$$

where:

$$\mathbf{Z} \triangleq \mathbf{R}^{-1/2} \mathbf{X} = \mathbf{R}^{-1/2} \mathbf{A} \mathbf{p} + \mathbf{R}^{-1/2} \mathbf{W} \rightarrow \mathbf{Z} \in \mathcal{N}(\mathbf{A} \mathbf{R}^{-1/2} \mathbf{p}, \mathbf{I}) \equiv \mathcal{N}(\mathbf{A} \mathbf{q}, \mathbf{I})$$

$\mathbf{q} \triangleq \mathbf{R}^{-1/2} \mathbf{p}$  is the "whitened" signal vector





- To implement the **Whitening Filter (WF)** we need to know the disturbance covariance matrix  $\mathbf{R}$ .
- If it is unknown, we should estimate it from the observed data.



## **Estimation of the Mean Vector and Covariance Matrix of a Random Vector**



- We need to collect  $K$  (possibly independent) realizations of the random vector  $\mathbf{X}$ :

$$\{\mathbf{X}_k\}_{k=1}^K \text{ IID} \rightarrow E\{\mathbf{X}_k\} = \boldsymbol{\eta}, \quad E\{(\mathbf{X}_k - \boldsymbol{\eta})(\mathbf{X}_k - \boldsymbol{\eta})^H\} = \mathbf{R}$$

- One possible estimator for the mean vector, whatever is the data pdf, is the **Sample Mean Vector (SMV)**, given by:

$$\boldsymbol{\eta} = E\{\mathbf{X}\} \rightarrow \hat{\boldsymbol{\eta}} = \frac{1}{K} \sum_{k=1}^K \mathbf{X}_k$$

- This estimate is **unbiased**:

$$E\{\hat{\boldsymbol{\eta}}\} = \frac{1}{K} \sum_{k=1}^K E\{\mathbf{X}_k\} = \frac{1}{K} \sum_{k=1}^K \boldsymbol{\eta} = \boldsymbol{\eta}$$

- This estimate is **consistent**:

$$\begin{aligned}
 \mathbf{C}_{\hat{\boldsymbol{\eta}}} &\triangleq E\{(\hat{\boldsymbol{\eta}} - \boldsymbol{\eta})(\hat{\boldsymbol{\eta}} - \boldsymbol{\eta})^H\} = E\left\{\left(\frac{1}{K} \sum_{k=1}^K \mathbf{X}_k - \boldsymbol{\eta}\right)\left(\frac{1}{K} \sum_{i=1}^K \mathbf{X}_i - \boldsymbol{\eta}\right)^H\right\} \\
 &= E\left\{\left(\frac{1}{K} \sum_{k=1}^K (\mathbf{X}_k - \boldsymbol{\eta})\right)\left(\frac{1}{K} \sum_{i=1}^K (\mathbf{X}_i - \boldsymbol{\eta})^H\right)\right\} \\
 &= \frac{1}{K^2} \sum_{k=1}^K \sum_{i=1}^K E\{(\mathbf{X}_k - \boldsymbol{\eta})(\mathbf{X}_i - \boldsymbol{\eta})^H\} \\
 &= \frac{1}{K^2} \sum_{k=1}^K \sum_{i=1}^K \mathbf{R} \delta_{k,i} = \frac{1}{K^2} \sum_{k=1}^K \mathbf{R} = \frac{1}{K} \mathbf{R} \xrightarrow{K \rightarrow \infty} \mathbf{0}
 \end{aligned}$$

$\lim_{K \rightarrow \infty} \hat{\boldsymbol{\eta}} = \boldsymbol{\eta}$  (convergence in mean square sense)

$$i.e. \quad \lim_{K \rightarrow \infty} E\{|\eta_i - \hat{\eta}_i|^2\} = 0, \quad 1 \leq i \leq N$$

- The **Sample Covariance Matrix (SCM)** estimator:

$$\mathbf{R} = E\{(\mathbf{X} - \boldsymbol{\eta})(\mathbf{X} - \boldsymbol{\eta})^H\} \rightarrow \hat{\mathbf{R}} = \frac{1}{K} \sum_{k=1}^K (\mathbf{X}_k - \hat{\boldsymbol{\eta}})(\mathbf{X}_k - \hat{\boldsymbol{\eta}})^H$$

- If we want the SCM to be full rank, i.e. invertible, we need  $K > N$  secondary data vectors.
- This estimate is **asymptotically unbiased** (unbiased if we know the mean vector, e.g. it is zero:  $\boldsymbol{\eta} = \mathbf{0}$ , as for thermal noise or radar clutter) and **consistent**.

$$\boldsymbol{\eta} = \mathbf{0} \Rightarrow E\{\hat{\mathbf{R}}\} = \frac{1}{K} \sum_{k=1}^K E\{\mathbf{X}_k \mathbf{X}_k^H\} = \frac{1}{K} \sum_{k=1}^K \mathbf{R} = \mathbf{R}$$

$$\lim_{K \rightarrow \infty} \hat{\mathbf{R}} = \mathbf{R} \quad (\text{convergence in mean square sense})$$

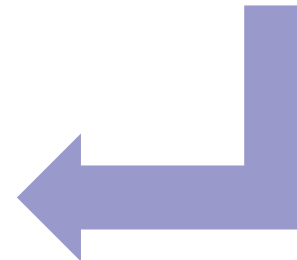
$$\text{i.e.} \quad \lim_{K \rightarrow \infty} E\left\{\left\|\mathbf{R}_{i,k} - \hat{\mathbf{R}}_{i,k}\right\|_F^2\right\} = 0, \quad 1 \leq i, k \leq N$$

- The **Sample Mean Vector** and the **Sample Covariance Matrix** estimators can be used whatever is the data distribution.
- It can be proved that they are also the **Maximum Likelihood (ML)** estimators if the data vector is Gaussian distributed:

$$(\hat{\boldsymbol{\eta}}_{ML}, \hat{\mathbf{R}}_{ML}) = \arg \max_{\boldsymbol{\eta}, \mathbf{R}} f_{\mathbf{X}}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_K; \boldsymbol{\eta}, \mathbf{R}) = \arg \max_{\boldsymbol{\eta}, \mathbf{R}} \prod_{k=1}^K f_{\mathbf{x}_k}(\mathbf{x}_k; \boldsymbol{\eta}, \mathbf{R})$$

$$\hat{\boldsymbol{\eta}}_{ML} = \frac{1}{K} \sum_{k=1}^K \mathbf{X}_k$$

$$\hat{\mathbf{R}}_{ML} = \frac{1}{K} \sum_{k=1}^K (\mathbf{X}_k - \hat{\boldsymbol{\eta}}_{ML})(\mathbf{X}_k - \hat{\boldsymbol{\eta}}_{ML})^H$$



- These estimators are **asymptotically unbiased**, **asymptotically efficient**, and **consistent**.
- They are usually used for **adaptive detection in Gaussian disturbance**.

- The multidimensional probability density function (pdf) of a real correlated Gaussian random vector is given by:

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^N |\mathbf{R}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\eta})^T \mathbf{R}^{-1}(\mathbf{x} - \boldsymbol{\eta})\right)$$

*Nx1 data vector*

$|\mathbf{R}| = \det\{\mathbf{R}\}, \quad \mathbf{R} = \mathbf{R}^T \rightarrow$  symmetric matrix

$$\mathbf{X} \in \mathcal{N}(\boldsymbol{\eta}, \mathbf{R}), \quad \boldsymbol{\eta} \triangleq E\{\mathbf{X}\} \quad \text{Mean Vector}$$

$$\mathbf{R} \triangleq E\{(\mathbf{X} - \boldsymbol{\eta})(\mathbf{X} - \boldsymbol{\eta})^T\} \quad \text{Covariance Matrix}$$

$$R_{i,j} \triangleq [\mathbf{R}]_{i,j} = \text{cov}\{X_i, X_j\} = E\{(X_i - \eta_i)(X_j - \eta_j)\}$$

- The multidimensional pdf of a complex correlated Gaussian random vector, with circular symmetry, is given by:

$$f_{\mathbf{x}}(\mathbf{x}) = \frac{1}{\pi^N |\mathbf{R}|} \exp\left(-(\mathbf{x} - \boldsymbol{\eta})^H \mathbf{R}^{-1} (\mathbf{x} - \boldsymbol{\eta})\right)$$

*Nx1 data vector*

$|\mathbf{R}| = \det\{\mathbf{R}\}, \quad \mathbf{R} = \mathbf{R}^H \rightarrow \text{Hermitian matrix}$

$$\mathbf{X} \in \mathcal{CN}(\boldsymbol{\eta}, \mathbf{R}), \quad \boldsymbol{\eta} \triangleq E\{\mathbf{X}\} \quad \text{Mean Vector}$$

$$\mathbf{R} \triangleq E\{(\mathbf{X} - \boldsymbol{\eta})(\mathbf{X} - \boldsymbol{\eta})^H\} \quad \text{Covariance Matrix}$$

$$R_{i,j} \triangleq [\mathbf{R}]_{i,j} = \text{cov}\{X_i, X_j\} = E\{(X_i - \eta_i)(X_j - \eta_j)^*\}$$

- **Problem:** Detect the presence of a signal of known shape and unknown complex amplitude  $A$  in additive correlated complex-valued Gaussian noise:

$$\begin{cases} \mathbf{X} = \mathbf{W} & H_0 : \text{SOI absent} \\ \mathbf{X} = A\mathbf{p} + \mathbf{W} & H_1 : \text{SOI present} \end{cases}$$

- The detection is a **binary hypothesis problem** whereby we decide either hypothesis  $H_0$  is true and only noise is present (decision  $D_0$ ) or hypothesis  $H_1$  is true and a signal of interest of known shape is present with noise (decision  $D_1$ ).
- **Solution:** we compare a function of the ML estimation of  $A$  with a threshold  $\lambda$   
 → **Adaptive Whitening Matched Filter (AMF):**

$$\hat{A}_{ML} = \frac{\mathbf{p}^H \hat{\mathbf{R}}^{-1} \mathbf{X}}{\mathbf{p}^H \hat{\mathbf{R}}^{-1} \mathbf{p}} \Rightarrow \Lambda_{AMF}(\mathbf{X}) = g(\hat{A}_{ML}) = \frac{|\mathbf{p}^H \hat{\mathbf{R}}^{-1} \mathbf{X}|^2}{\mathbf{p}^H \hat{\mathbf{R}}^{-1} \mathbf{p}} \underset{H_0}{\overset{H_1}{>}} \lambda$$

- The inverse of the **Sample Covariance Matrix (SCM)** can be efficiently calculated through the data matrix **Singular Value Decomposition (SVD)**:

$$\boldsymbol{\eta} = \mathbf{0}, \quad \hat{\mathbf{R}} = \frac{1}{K} \sum_{k=1}^K \mathbf{X}_k \mathbf{X}_k^H$$

- The data matrix  $\mathbf{X}_M$  is the matrix which contains the  $K$  data vectors:

$$\mathbf{X}_M \triangleq \begin{bmatrix} \mathbf{X}_1 & \mathbf{X}_2 & \cdots & \mathbf{X}_K \end{bmatrix}, \quad K \geq N \text{ (usually } 2N \leq K \leq 3N)$$

$N \times K$

$$\hat{\mathbf{R}} = \frac{1}{K} \sum_{k=1}^K \mathbf{X}_k \mathbf{X}_k^H = \frac{1}{K} \mathbf{X}_M \mathbf{X}_M^H$$

$N \times N$

- The data matrix  $\mathbf{X}_M$  is full rank, i.e.  $\text{rank}(\mathbf{X}_M) = N$ , with probability 1 if  $K \geq N$ .



- The inverse of the **Sample Covariance Matrix (SCM)** can be efficiently calculated through the data matrix **Singular Value Decomposition (SVD)**:

$$\mathbf{X}_M = \underset{N \times K}{\mathbf{U}} \underset{N \times N}{\mathbf{\Sigma}} \underset{N \times K}{\mathbf{V}^H} = \underset{N \times N}{\mathbf{U}} \begin{bmatrix} \underset{N \times N}{\mathbf{\Sigma}_1} & \underset{N \times (K-N)}{0} \end{bmatrix} \begin{bmatrix} \underset{N \times K}{\mathbf{V}_1^H} \\ \underset{N \times (K-N)}{\mathbf{V}_2^H} \end{bmatrix} = \underset{N \times N}{\mathbf{U}} \underset{N \times N}{\mathbf{\Sigma}_1} \underset{N \times K}{\mathbf{V}_1^H}$$

$$\begin{aligned} \hat{\mathbf{R}} &= \frac{1}{K} \mathbf{X}_M \mathbf{X}_M^H = \frac{1}{K} \underset{N \times N}{\mathbf{U}} \underset{N \times N}{\mathbf{\Sigma}_1} \underset{N \times K}{\mathbf{V}_1^H} \left( \underset{N \times N}{\mathbf{U}} \underset{N \times N}{\mathbf{\Sigma}_1} \underset{N \times K}{\mathbf{V}_1^H} \right)^H \\ &= \frac{1}{K} \underset{N \times N}{\mathbf{U}} \underset{N \times N}{\mathbf{\Sigma}_1} \underset{N \times K}{\mathbf{V}_1^H} \underset{K \times N}{\mathbf{V}_1} \underset{N \times N}{\mathbf{\Sigma}_1^H} \underset{N \times N}{\mathbf{U}^H} = \frac{1}{K} \underset{N \times N}{\mathbf{U}} \underset{N \times N}{\mathbf{\Sigma}_1^2} \underset{N \times N}{\mathbf{U}^H} \end{aligned}$$

$$\hat{\mathbf{R}}^{-1} = K \left( \underset{N \times N}{\mathbf{U}} \underset{N \times N}{\mathbf{\Sigma}_1^2} \underset{N \times N}{\mathbf{U}^H} \right)^{-1} = K \left( \underset{N \times N}{\mathbf{U}^H} \right)^{-1} \underset{N \times N}{\mathbf{\Sigma}_1^{-2}} \underset{N \times N}{\mathbf{U}^{-1}} = K \underset{N \times N}{\mathbf{U}} \underset{N \times N}{\mathbf{\Sigma}_1^{-2}} \underset{N \times N}{\mathbf{U}^H}$$

- which can be easily calculated, since the matrix  $\mathbf{\Sigma}_1$  of singular values is diagonal.

$$\begin{aligned}\Lambda_{AMF}(\mathbf{X}) &= \frac{|\mathbf{p}^H \hat{\mathbf{R}}^{-1} \mathbf{X}|^2}{\mathbf{p}^H \hat{\mathbf{R}}^{-1} \mathbf{p}} = \frac{|\mathbf{p}^H (K \mathbf{U} \boldsymbol{\Sigma}_1^{-2} \mathbf{U}^H) \mathbf{X}|^2}{\mathbf{p}^H (K \mathbf{U} \boldsymbol{\Sigma}_1^{-2} \mathbf{U}^H) \mathbf{p}} = K \frac{|\mathbf{p}^H (\mathbf{U} \boldsymbol{\Sigma}_1^{-2} \mathbf{U}^H) \mathbf{X}|^2}{\mathbf{p}^H (\mathbf{U} \boldsymbol{\Sigma}_1^{-2} \mathbf{U}^H) \mathbf{p}} \\ &= K \frac{\left| (\mathbf{U}^H \mathbf{p})^H \boldsymbol{\Sigma}_1^{-2} (\mathbf{U}^H \mathbf{X}) \right|^2}{(\mathbf{U}^H \mathbf{p})^H \boldsymbol{\Sigma}_1^{-2} (\mathbf{U}^H \mathbf{p})} = K \frac{|\mathbf{q}^H \boldsymbol{\Sigma}_1^{-2} \mathbf{Z}|^2}{\mathbf{q}^H \boldsymbol{\Sigma}_1^{-2} \mathbf{q}} = K \frac{\left| \sum_{i=1}^N \frac{1}{\sigma_i^2} q_i^* z_i \right|^2}{\sum_{i=1}^N \frac{1}{\sigma_i^2} |q_i|^2} \underset{H_0}{\overset{H_1}{>}} \lambda\end{aligned}$$

$\mathbf{Z} = \mathbf{U}^H \mathbf{X}$  [DKLT (decorrelating transformation) using the estimated eigenvector matrix  $\mathbf{U}$ ]

$\mathbf{q} = \mathbf{U}^H \mathbf{p}$  [transformed signal vector]